Social Hamiltonian and Legal Engineering

A Constructive Approach to the Latent Torus Concept via Quantum Algorithms and Smart Contracts

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Abstract

This study quantifies Large Language Models (LLMs) and humanoids as a new labor force and describes the transformation of economic structures brought about by "super-fluid task allocation involving humans," facilitated by tokenized task transactions built on blockchain technology, from the perspective of statistical physics. Furthermore, we devise a constructive approach called "Legal Engineering" and discuss its governance mechanisms.

First, we define the price fluctuations of tokenized tasks as "work volatility." Within the scope where specific assumptions (information friction, interaction amplification, and approximate effective temperature) hold, phase-transition-like behaviors can occur. Volatility is interpreted as an operational approximation of "social temperature" emerging from amplified information friction and interactions.

As a governance mechanism to suppress this entropy increase, we propose the "Latent Torus," an information event horizon. The Latent Torus handles internal optimization invisible from the frontend and ensures sustainable social order by recirculating only optimized parameters to smart contracts. Here, by combining quantum optimization with "Semantic Intervention" via "Regulated LLMs," we aim for stabilization based on "semantic depth" rather than apparent liquidity.

Furthermore, we propose a "Grand Unified Algorithm" to simultaneously handle economic efficiency (Hamiltonian minimization), humanity (Well-being), and social credit (Proof of Trust) within a single mathematical framework. The scope of this paper is not to advocate for immediate control of society as a whole, but rather to provide a conceptual model for optimizing and auditing trade-offs between indicators in a consistent manner under limited task spaces, participant sets, and operational rules. As a concrete model, we present "Computational Social Contract Theory (CSCT)" and confirm its behavior and limitations under various assumptions through quantitative analysis using multi-agent simulations.

This theory presents a design policy for realizing "verifiable concealment" in governance under certain assumptions (circuitability, computational security, and sound key management) using cryptographic techniques such as zero-knowledge proofs (zk-SNARKs). We explore the possibility of hiding internal optimization details while maintaining compliance with the Constitutional Core, allowing citizens to verify legitimacy.

This paper presents a conceptual proposal for institutional design in a post-capitalist society and examines the redesignability of money and law. Note that the quantitative results of this paper are positioned as exploratory simulations and do not directly claim predictive confirmation.

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1 Introduction

1.1 Background: Democratization of Tasks and Entropy Increase

In his book "The Future of Work" (Malone 2004), Thomas W. Malone foresaw how falling communication costs would transform business organizational structures. Specifically, as information transmission becomes cheaper, market-based decentralized coordination becomes relatively more advantageous, and some of the coordination functions previously handled by hierarchies can be externalized. However, even as communication becomes cheaper, as long as costs associated with the **verification and enforcement** of contracts (who did what, whether promises were kept, and how to settle in case of violations) remain, hierarchies will continue to be devices for lowering transaction costs. The development of blockchain technology (Nakamoto 2008) and smart contracts can externalize this previously unresolved trust/verification cost as a protocol, thus challenging the assumption of the superiority of centralized hierarchical structures.

Currently, we are in the midst of a transformation described as the "democratization of tasks." We are entering an era where anyone can cut out labor (tasks) as tokens at any granularity and trade them across borders.

However, the problem addressed in this paper is not the immediate equation of "freedom" with "disorder." "Disorder" here refers to the instability of state estimation from the perspective of an external observer because, although the degrees of freedom allowed by the system have increased, the **process of imposing constraints** (observation, verification, and enforcement) cannot keep up. In other words, information available at the time of decision-making is always delayed, and because unresolved elements (unverified, unsettled, or unresolved disputes) remain, the number of possible states (state candidates) that can consistently explain the same observation increases.

In this paper, "entropy increase" is defined as a state where, due to the high frequency and fine granularity of the task market, the **update rate of the market state** exceeds the reaction rate of coordination processes such as **agent cognition, verification, settlement, and dispute resolution**, resulting in an increase in **uncertainty (information entropy) for the observer** (Time-scale Mismatch).

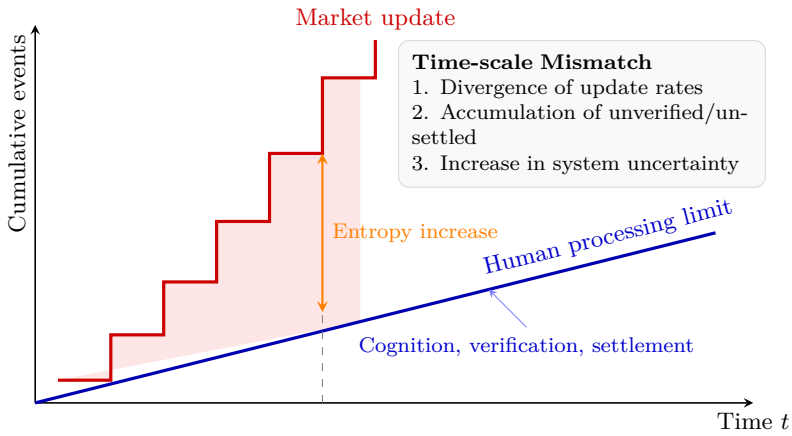

Figure 1: Conceptual diagram of entropy (uncertainty) increase due to time-scale mismatch.

Specifically, we use (i) information search costs, (ii) coordination failure probabilities such as contract non-performance or verification delays, (iii) reputation evaluation variance, and (iv) price series instability as observable proxy variables.

When this time-scale mismatch exceeds a critical point, the price mechanism may cease to integrate distributed information and instead lead to self-amplification (bubble/crash) through delayed feedback (Minsky

2008). In this paper, we treat this instability as "work volatility," defined later.

This study presents a framework with explicit verifiable assumptions and limitations for the socio-engineering challenge of how to convert "disorder born of freedom" into a new "order (negentropy)."

1.2 Problem Statement: Limits of Fiat Currency and Physical Reinterpretation of Law

The essence of the crisis we face is that the existing social OS—namely, the "payment rails centered on fiat currency" and "static statutory law"—is failing to keep up with the speed, granularity, and complexity of transactions in digital space, both in terms of technical interfaces and operational processes.

1. **Bottleneck of Currency (Credit):** In microtask markets, even if each task is paid for with immediate settlement (cash), **task execution uncertainty** (quality, deadline, non-performance) and **verification/dispute resolution delays** remain. Therefore, participants must practically rely on **dynamic credit** such as credit (payment, advance, guarantee) and reputation (execution capability). In this paper, we broadly treat this "credit" as a **state variable** that includes not only payment ability but also execution ability and compliance.

   As automated execution by algorithms becomes commonplace, "credit" tends to be handled more formally (protocol-wise). However, the problem is not just the "technical interface" but the **credit granularity mismatch** caused by the **centralized and coarse-grained processes of credit creation and evaluation**. That is, for ultra-high-speed and fine-grained transactions, (a) credit, audit, and termination decisions are made at institutionally coarse time intervals, (b) credit (limits, collateral, sanctions) cannot be directly and conditionally controlled from code, and (c) as a result, the update of credit lags behind market updates.

   Of course, while value transfer interfaces may improve through stablecoins or CBDCs, this alone does not guarantee the redesign of evaluation criteria—**"what to consider as credit (which state variables to connect to collateral/sanctions)"**—and their fine-grained execution. This paper treats this point as a redesign problem of the social OS, including currency (credit). Technical constraints also coexist, such as (i) non-linearity of fees in micropayments, (ii) latency in cross-border settlements, (iii) granularity of settlement finality and auditability, and (iv) direct controllability of conditional payments (escrow/penalty) from code.

2. **Lag of Law:** While technological and social changes accelerate, human legislation and operation take time at each stage of institutional design, consensus building, and enforcement. This "speed mismatch" creates regulatory gaps or over-regulation, which can result in market instability and the amplification of external diseconomies.

Here, we need to extend jurisprudence from a "science of interpretation" to a "science of implementation." What this research proposes as **"Legal Engineering"** is an attempt to design institutions so that formalizable parts of **norms (ought)** are extracted and implemented as **automatically executable constraints and procedures (code)**, ensuring that violations and exceptions do not get buried as operational ambiguities but instead appear as **observable consequences** (see section 2.1 for details on definition and stance).

On the other hand, converting natural language law into code involves the cost of fixing interpretations (loss of flexibility). This paper aims for implementation not as "mere automation," but to bring enforcement costs as close to zero as possible and to realize **micro-justice** even for minute transactions and microtasks. Therefore, in this study, assuming the risk of rigidity due to implementation, a two-layer design including low-frequency safety valves such as Human Veto is positioned as a governance requirement.

1.3 Objective: Building a Grand Unified Algorithm for Institutional Design

The objective of this paper is to analyze price fluctuations (work volatility) in a tokenized economy using statistical physics methods (Kadowaki and Nishimori 1998), and to define governance requirements for suppressing self-amplification.

To this end, what this paper considers as the "object to be controlled" is not price itself, but mechanisms such as (i) coordination delays caused by the latency of transactions, verification, and settlement, (ii) variance in credit evaluation among heterogeneous participants, and (iii) instability caused by cascades of constraint violations (non-performance, etc.). Therefore, the necessary requirements are at least: (a) the ability to iterate optimization and verification internally at high speed, (b) not exposing internal states (optimization details) to the outside, and (c) yet ensuring that **the outcome being constraint-compliant is verifiable** (this paper positions this as a requirement similar to an "information event horizon").

As a model that meets these requirements, this study proposes the governance mechanism **"Latent Torus."** Here, "Latent" refers to holding internal states as latent variables while projecting only verifiable outputs to the outside. Furthermore, in this study, we treat the two cycles (phases) of high-frequency market updates and low-frequency institutional updates as a direct product, expressing that phase space as $S^1 \times S^1$ and mapping the double circulation into an **analyzable coordinate system** (details in section 2 and section 3).

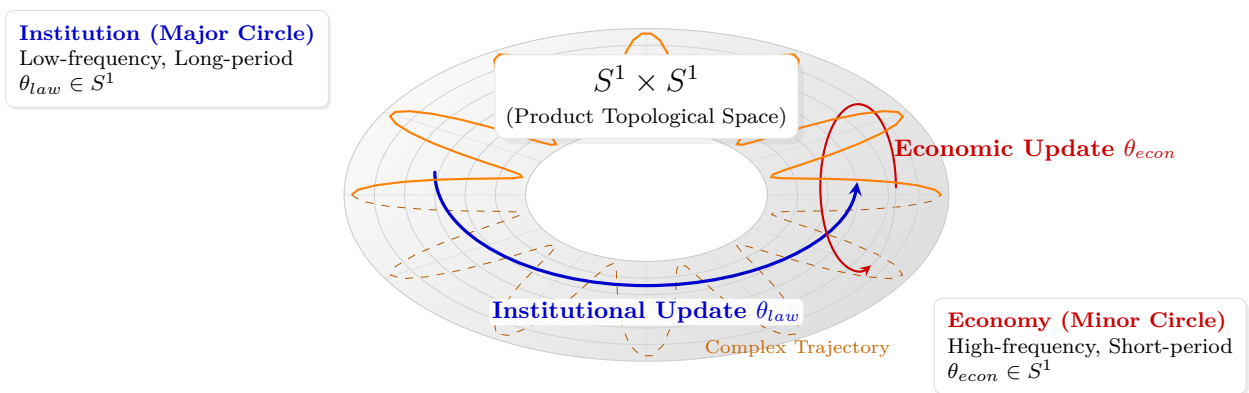

Figure 2: Phase space of the Latent Torus: Double circulation structure of institution and economy.

In particular, in addition to the "economic cost (H_{econ})" that traditional economics has handled, we map "social friction" handled by ethics and law as **constraint violation penalties (or costs of external diseconomies)** onto H_{social}. We try to treat both as a single objective function (analogue of free energy) with weighting, and mathematically build a "Grand Unified Algorithm" to minimize it.

Furthermore, this study eventually attempts to introduce a "Grand Unified Theory" that treats law, economy, and human happiness (Well-being) integrally. As a concrete approach, we present a conceptual model called **"Computational Social Contract Theory (CSCT)"** and verify its feasibility and sustainability through quantitative demonstration using multi-agent simulations (Chapters 6 to 8).

1.3.1 Key Questions and Verification Strategy

The central questions of this research are as follows:

1. Under what conditions can price fluctuations (work volatility) diverge self-amplifyingly (bubble/crash) in a tokenized task market?

2. How does the design of information liquidity (semantic depth) and institutional constraints ($\lambda_{core}, \lambda_{policy}$) bring about trade-offs between volatility and well-being?
3. Can governance via "verifiable concealment (zk-SNARKs, etc.)" satisfy the requirements for transparency, auditability, and privacy simultaneously?

In this paper, we ensure verifiability by (i) mapping the theoretical model onto observable quantities (price series, logs, constraint violation rates, etc.), (ii) performing sensitivity analysis of design variables in simulations, and (iii) presenting falsifiable hypotheses (e.g., intervention frequency or information friction changing critical behavior).

The principal objective of this paper is to design a Social Hamiltonian H that represents the social state as an objective function. We construct a governance mechanism capable of improving well-being while suppressing work volatility by implementing state transitions that satisfy rigorous constraints through the Latent Torus framework.

1.4 Contributions

The contributions of this study are summarized in the following three points:

1. **Constructive Formulation of Social Hamiltonian**: Defined the social state as an objective function and redefined governance as constrained optimization explicitly including the Constitutional Core $\lambda_{core} \to \infty$.
2. **"Verifiable Concealment" via Latent Torus**: Presented a governance architecture that allows only the fact that outputs satisfy constraint fulfillment and cost reduction to be verifiable via zk-proofs while hiding internal optimization.
3. **Design of PoT (Proof of Trust) and Safety Valves**: Mapped credit (τ) decay and Sybil resistance onto a mathematical model, and showed design policies to suppress operational catastrophe risks through low-frequency safety valves such as Human Veto.

The structure of this paper is as follows. In section 2, we present the theoretical framework, and in section 3, we build the mathematical models for work volatility and the Latent Torus. In section 4 and section 5, we propose the Grand Unified Algorithm and the consensus foundation (PoT) and verify their engineering feasibility. The expected social impact and structural reforms are discussed in section 6, along with long-term simulation results. Finally, in section 8, we formulate Legal Engineering and CSCT, and state the conclusions in section 8.6.

2 Theoretical Framework

This chapter aims to **theoretically organize** the problem awareness and proposed concepts presented in section 1 based on previous research and definitions. While the introduction summarized concepts for the reader's entry, this chapter and beyond will clarify definitions, lineages, and differences, constituting a relationship of **"overview → grounds and positioning"** rather than duplication.

2.1 Stance of Legal Engineering

"Legal Engineering" in this paper is a constructive approach that views law not as **clauses to be interpreted** but as **formal specifications to be implemented**, designing, verifying, and updating the state transition rules of the social OS.

While traditional LegalTech mainly targets business efficiency (document creation, search, contract management, etc.), Legal Engineering places the following at its core:

- **Formalization of the Constitutional Core (Layer 0/1)**: Explicitly stating constraints such as human rights and due process, and excluding violations from the state space.
- **Governance as Constrained Optimization**: Treating the design and updating of the Social Hamiltonian H as a mechanism design problem.
- **Verifiability (zk)**: Making only the satisfaction of constraints and the legitimacy of state transitions verifiable while hiding internal optimization.

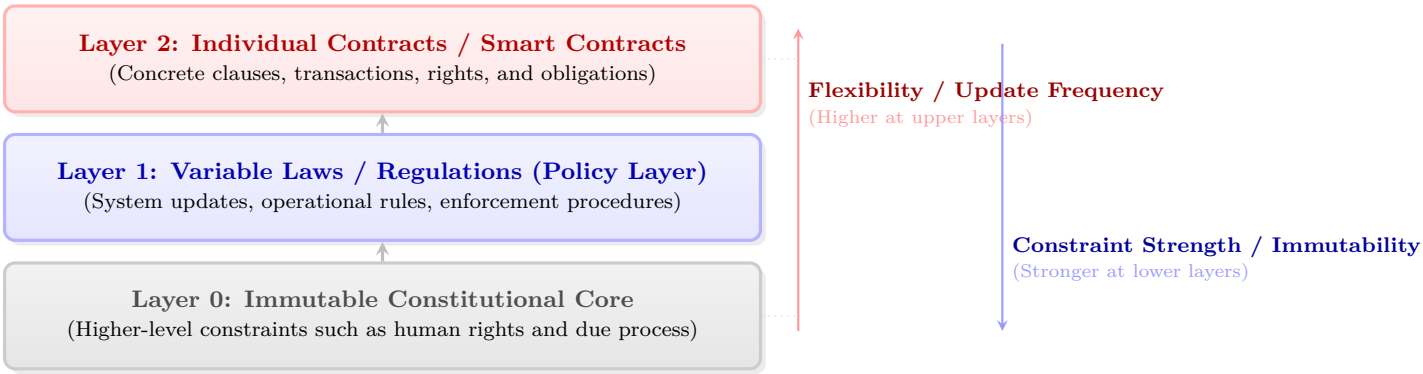

Figure 3: Hierarchical structure of law (Layer 0/1/2) and its correspondence with layer concepts in blockchain.

With this redefinition, the design challenge of the legal system shifts from "interpretation of clauses" to "implementation that withstands ambiguity of specifications, oracle pollution, and semantic attacks."

Implementation Requirement: Provable LLM Computing: Recent implementations in the "Provable Computer" family have proposed designs that trace the LLM inference process itself at the instruction level and present cryptographic proofs consistent with the inference results to an external verifier (Abdel-Stark 2026). This trend is important for connecting **verifiable concealment** in the Legal Engineering of this paper from an abstract ideal to an implementation requirement.

In the context of this study, the following three points are particularly suggestive:

- **Enforcement of Reproducibility**: Explicitly stating a deterministic inference path to obtain the same output from the same input, same model weights, and same quantization settings, making it a requirement that "the same calculation can be re-executed" during auditing.
- **Separation of Verification Responsibilities**: Delegating heavy proof generation (proving) to off-chain computing resources and performing only lightweight verification (verifying) on the on-chain/public side.
- **Balance between Model Secrecy and Public Audit**: Allowing third parties to verify the fact that "the calculation was performed according to the given specifications" without disclosing all weights or intermediate representations inside the model.

Therefore, regulated LLMs in Legal Engineering must be designed not only as "high-precision advisors" but also as **provable state transition functions**. This means extending the legitimacy substrate for operating law as formal specifications from an explainability-centered (XAI) framework to a verifiability-centered framework.

2.2 Democratization of Tasks and Tokenomics

Various experiments with proprietary or local currencies have been conducted by regions and companies in the past. While most were aimed at economic revitalization or community strengthening, the "tokenization of tasks" using blockchain technology is distinct. Here, work (tasks) is carved out as a single package, value is assigned in the form of tokens, and it is traded in the market.

This fundamentally transforms the flow of task progression. Instead of task assignment within a traditional corporate organization (top-down), it shifts to a matching and consensus-building process through market mechanisms (bottom-up). This means that tasks themselves have liquidity like financial products, and prices fluctuate according to time and supply/demand.

2.3 Organizational Limits and the Calculation Debate: An Austrian View

As the democratization of tasks progresses, how will the role of corporate organizations change? The economic calculation debate of the Austrian school provides important suggestions for this question. According to (Klein 1996), Mises and Rothbard argued that if an organization (firm) becomes too large and internal transactions swallow up all intermediate goods markets, it becomes impossible to refer to external market prices, making rational economic calculation impossible.

"Transfer prices" within a firm are merely imitations of market prices and do not reflect true scarcity. The tokenization of tasks can be interpreted as an attempt to assign an external market price (token price) to each internal task that was previously invisible inside the "company" black box. This enables rational calculation based on market principles even for intra-company tasks, opening a path to resolve inefficiencies caused by organizational bloat.

2.4 Cybernetic Pioneers: Failure of Planned Economy and Feedback

While the Austrian school pointed out the impossibility of calculation, attempts were made to control society through an engineering approach in the field of cybernetics. Stafford Beer designed a system to control the national economy in real-time in Chile's "Project Cybersyn" (Beer 1966; Medina 2011). However, due to the limits of communication technology and computing power at the time, delays (latency) in the feedback loop could not be resolved, and combined with political turmoil, this attempt failed to leave sustainable results.

Cybernetics, proposed by Norbert Wiener, is based on feedback control that minimizes the difference (error) between the input and output of a system (Wiener 1948). The "Latent Torus" proposed in this research is an attempt to revive real-time economic control once dreamed by Beer in the modern era by incorporating "future prediction (feed-forward)" using quantum computing and "immutable state management" using blockchain into this classical feedback loop.

2.5 Transaction Costs and Smart Contracts

On the other hand, market transactions involve transaction costs (costs of search, negotiation, and enforcement). According to (Williamson 1981)'s transaction cost theory, transactions with high asset specificity and high uncertainty are more efficient in internal organizations (hierarchies) than in markets.

However, the advent of smart contracts may overturn this premise. Automating the formation and enforcement of contracts through programs dramatically reduces the costs associated with market transactions. In particular, the automation of task completion confirmation and reward payment expands the scope of market mechanisms to minute task areas that could previously only be handled within internal organizations. Furthermore, as shown by (Haffar and Özceylan 2024), blockchain-based reputation systems can

realize automation while ensuring transparency and sustainability even in complex processes such as supplier selection.

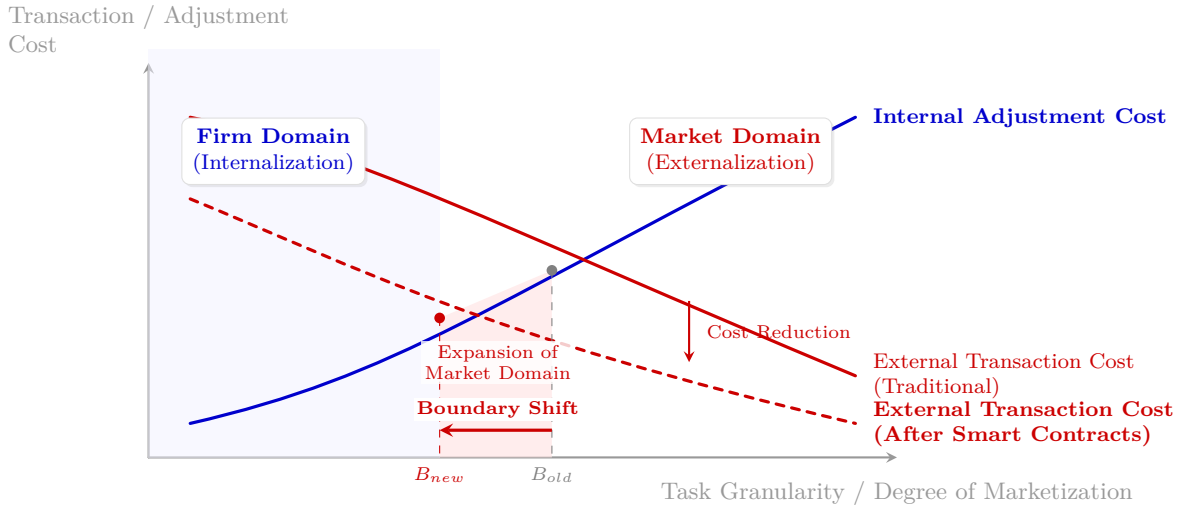

Figure 4: Shift of boundary between Organization (Firm) and Market due to falling transaction costs (Smart Contracts).

The simultaneous progression of this "minimization of transaction costs" and "maximization of calculability" forms the theoretical basis for why the boundary between organization and market becomes more fluid than ever before, generalising the "cross-border phenomenon" proposed in this study.

However, lowering transaction costs does not always mean an improvement in well-being. As tasks become more subdivided and fluid, price discovery accelerates, but information asymmetry and evaluation noise increase, potentially causing volatility to self-amplify. This paper formalizes this point as the relationship between work volatility V and information liquidity $\mathcal{L}_{info}$, explicitly stating the condition that "further marketization can lead to instability."

2.6 Intra-company Currency and Gamification: DISCO as a Precursor

An early form of the "democratization of tasks" discussed in this research can be found in DISCO Corporation's "Will Accounting System" (DISCO Corporation 2023). In this company, internal currency called "Will" is used to quantify the value of all internal resources and labor, from the use of meeting rooms to the undertaking of microtasks, and transactions are conducted in an auction format. This is an attempt to introduce a pseudo-market mechanism inside the organization and encourage individual autonomous decision-making (bidding for "jobs they want to do").

Such systems can increase employee engagement by gamifying business processes. However, they also involve psychological challenges. As pointed out by Deci (Deci 1971) and Frey (Frey 1997), there is a risk of a "crowding out effect" occurring, where extrinsic rewards (money or tokens) actually diminish motivation for actions previously performed based on intrinsic motivation (enjoyment of the work itself).

The tokenization of tasks in this study is not merely a strengthening of financial incentives, but also considers the possibility of avoiding this crowding out and creating a new motivation structure (crowding in) by combining the "absoluteness of contracts" via smart contracts with "semantic intervention" via AI.

2.7 Epistemological Stance: Social Physics and Thermodynamics

The approach of this study stands on the same epistemological footing as Black and Scholes' introduction of thermodynamics (heat equation) into option pricing theory (Black and Scholes 1973). That is, it discards the semantic interpretation (who intended what) of individual "tasks" and "laws" and focuses solely on their statistical properties as "signals" recorded on the blockchain. Thus, it attempts to reimplement a Hobbesian "Leviathan (governance mechanism)" (Hobbes 1651) not as a political authority, but as a thermodynamic "homeostatic mechanism."

This perspective is in line with the "Social Physics" approach proposed by Alex Pentland (MIT) (Pentland 2014). Pentland viewed the transmission of human behavior and ideas as physical "flows" and sought to predict and control social phenomena through big data analysis. This study applies this "physics of society" to the "immutable laboratory" of blockchain, extending it from mere observation to the realm of "engineering control (Legal Engineering)."

2.8 Contributions and Differences from Previous Research

In this section, we delineate the specific contributions and novelty of this study relative to the existing literature.

Furthermore, the Proof of Trust (PoT) proposed in this research can be connected to the concept of "Verifiable Credential" (**W3C2022**). By mapping trust onto a Social Hamiltonian, we can achieve a highly scalable consensus mechanism.

On the other hand, it has been organized that graph-based trust evaluation in OSNs faces common difficulties such as (a) path dependence, (b) trust decay, (c) opinion conflict, and (d) attack resistance (Sybil, etc.) (Jiang, Zhang, and Ong 2016). This paper reinterprets these as design requirements: (i) defining PoT as a non-transferable and updatable dynamic quantity, (ii) incorporating damping/semantic annealing into consensus algorithms, and (iii) closing attack surfaces operationally through oracle fallbacks and auditing.

2.8.1 Contributions

- **Unified Description of Social Costs**: Formulating economic costs and social friction (ethics/law) uniformly as a single Hamiltonian H and free energy minimization.
- **Engineering Definition of Latent Torus**: Defining the optimization core as an "information event horizon," making only the legitimacy of outputs verifiable while hiding internal states (verifiable concealment).
- **Stratification of Law and Recursive Verification**: Providing an architecture to recursively verify state transitions and law updates within the same framework, assuming a Layer 0/1/2 legal system.
- **Unified Processing of Heterogeneous Agents**: Projecting humans, LLMs, and robots onto the same state space with a general-purpose labor tensor, handling governance and market design with a consistent model.
- **Stabilization Mechanism for Consensus**: Proposing damping (circuit breakers) to suppress the runaway of crowd psychology and semantic annealing for escaping local optima as algorithms.

2.8.2 Comparison with Previous Research (Summary)

Table 1 summarizes the positioning relative to major related domains (details handled in each section).

Table 1: Summary of Differences Between This Study and Previous Research

Domain	**Representative Approach**	**Typical Limitations**	**Differences in This Study**
Cybernetics (Cybersyn)	Real-time planning / feedback control	Comm/calc delay, political arbitrariness	Black-boxing optimization under zk-verifiable constraints
Mechanism Design	Incentive design (voting/tax/market)	Identity/Sybil resistance, implementation difficulty	Connection to implementation layer based on PoT (non-transferable trust)
Crypto Governance	Verification by proof / scaling	Ambiguity of what to prove (obj function/constitution)	Clarification of verification propositions assuming layer stratification (Layer 0/1/2)
SBT / Reputation	Non-transferable tokens / reputation propagation	Dynamics (decay/redistribution), connection to social goals	Treating τ as a dynamic physical quantity and integrating with H minimization

3 Model Construction: Socio-physical Approach and Latent Torus

3.1 Social Hamiltonian and Work Volatility

In a tokenized task market, individual workers and tasks do not exist independently but behave as an interacting many-body system. To understand the behavior of such complex systems, traditional microeconomic approaches alone are insufficient. Equilibrium analysis based on individual rationality has its limits in today's digital economy, where a vast number of agents interact in real-time to form emergent patterns.

Therefore, this study adopts the methods of "statistical physics," particularly "econophysics," which derives the macroscopic properties of systems with many interacting elements from microscopic laws. By viewing society as a whole as a giant physical system and describing its energy state (the aggregate of credit and value), we introduce a Hamiltonian H, similar to those used in spin glass theory. This is an attempt to physically formalize the "cost function" of society, explicitly describing what the components of society are unconsciously trying to optimize.

$$H = -\mu \left(\sum_{\langle i,j \rangle} J_{ij} \vec{\sigma}_i \cdot \vec{\sigma}_j + \sum_i \vec{h}_i \cdot \vec{\sigma}_i \right) + \sum_k \lambda_k \Phi_k(\{\vec{\sigma}\}) \tag{1}$$

3.1.1 Mapping the Social Hamiltonian (Variables, Social Concepts, and Proxy Indicators)

To connect Equation (1) to an empirically testable description rather than a "metaphor," we map the main variables to social concepts and observable proxy indicators.

Table 2: Mapping the Social Hamiltonian (Variables, Social Concepts, and Proxy Indicators)

Variable	Social Concept	Proxy Indicators (Example Definition)	Data Source / Window
$\vec{\sigma}_i$	Agent state	Create embedding $e_i(t)$ from task logs, evaluations, and skill history; normalize as $\vec{\sigma}_i(t) := e_i(t)/\|e_i(t)\|$	Work logs / eval history (W_σ)
J_{ij}	Co-op/comp	Bonds from co-occurrence of co-authorship, joint tasks, or mutual evaluations: $J_{ij} := \alpha\,\mathrm{corr}(y_i, y_j) + \beta\,\mathrm{cooccur}(i,j) - \gamma\,\mathrm{conflict}$	Joint work / transaction / evaluation graph (W_J)
$\vec{h}_i$	Norms, incentives	Policy vector $u(t)$ and text embedding $n(t)$ decomposed from taxes, penalties, and narratives: $\vec{h}_i(t) := W_u\,u(t) + W_n\,n(t)$	System parameters / announcement docs
Φ_k	Constraints	Implement as a constraint judgment function for state S: $\Phi_k(S) \in \{\mathrm{True}, \mathrm{False}\}$ or $\mathrm{margin}_k(S)$	Audit logs / formal verification results
λ_k	Weight (Hard/Soft)	λ_{core} as exclusion from search space (Hard), λ_{policy} as trade-off weight (Soft) reachable by governance	Protocol specs / weight ledger
μ	Market sensitivity	Estimate variation in V_t or price elasticity before and after intervention using state-space models	Price series + intervention history (W_μ)

3.1.2 Connection to Real-World Data: Minimum Log Spec and Estimation Pipeline

For the mapping in this section to have practical "measurability," it is necessary to fix at least (i) which events, (ii) at what granularity, (iii) with which identifiers (IDs) are joined, and (iv) across what window width they are aggregated. Here, assuming a general platform where a task market (task request, acceptance, delivery, evaluation, payment) exists, we specify the **Minimum Viable Logs** required by this model.

Minimum Viable Logs: While assuming the confidentiality of individual data (privacy protection), at least the following events are required with timestamps:

- **Task Ledger**: Task ID, creation time, required skills (tags), reward p_t, deadline, and requester ID (can be anonymized).
- **Work Log**: Acceptance ID, start/end time, deliverable hash, work time, tools used, and (if possible) references to task description $d(T)$ and execution log $\ell(T)$.
- **Evaluation Log**: Evaluator ID (can be anonymized), evaluation score, comment (text), dispute flag, and evaluation confirmation time.
- **Interaction Graph**: Joint work (co-authorship on the same task), review relationships, and directed edges of mutual evaluation (weights are frequency or score).
- **Policy/Protocol Log**: Policy parameters $u(t)$ such as tax rates, subsidies, and penalties; announcement (narrative) text, application start time, and version ID.
- **Audit/Constraint Log**: Judgment results of constraints Φ_k (True/False or margin), violation categories, fallback triggers, and history of freezes/rollbacks.

Identification Units (Key Design): To enable estimation, at least (i) Task ID, (ii) Agent ID (subject ID including humans/LLMs/robots), and (iii) Policy Version ID must be consistent across all logs (anonymization is possible, but **traceability of the same subject** is necessary).

Estimation Pipeline (Outline of Operational Procedures):

1. **Preprocessing**: Remove missing or outlier data (abnormal rewards, duplicate evaluations, etc.) and align events related to the same task chronologically.
2. **Feature Extraction**: Estimate $\mathcal{L}_{info}(T)$ from $d(T)$ and $\ell(T)$, and unify evaluations, deliverables, and deadline deviations into a standard schema.
3. **Interaction Estimation**: Estimate $\hat{J}_{ij}$ from the Interaction Graph (allowing dimensionality reduction through sparse regularization or clustering).
4. **Intervention Response Estimation**: Estimate the variation of μ (intervention effect) using policy/protocol update points.
5. **Verification**: Output confidence intervals and sensitivity analysis of estimated values and confirm that conclusions do not depend on specific assumptions.

The above Minimum Viable Logs are a **necessary condition** for connecting the variables in this paper to their **observational/operational definitions**. That is, satisfying these requirements allows the variables in this paper to be treated as **objects of observation, estimation, and audit** rather than "metaphors" (however, being estimable does not immediately mean being **uniquely identifiable**).

To avoid ambiguity during implementation, this paper fixes the roles of primary symbols as shown in table 3 (one symbol – one role).

Table 3: Notation Role Map (for Implementation)

Symbol	Role
$\vec{\sigma}_i$	Agent state vector (Physical model side)
c_i	Legal claim vector (Dispute optimization side)
u_i	Citizen input signal (Governance interface side)
τ_i	Trust potential (Reputation / Voting weight)
$t_{breaker}, t_{market}, t_{LLM}, t_{relax}$	Time constants for control and intervention

Here, $\vec{\sigma}_i$ is the normalized state vector ($|\vec{\sigma}_i| = 1$) of each agent[1]. Additionally, the weight λ_k in the constraint potential Φ_k of the third term is implemented as $\lambda_{core} \to \infty$ (Hard Constraint / Satisfiability Problem) for "Constitutional Core" involving basic human rights and physical laws, and as a finite λ_{policy} (Soft Constraint / Penalty Term) for other policy-based guidance. This mathematically eliminates the risk of human rights violations being justified as "numerical compromises" in the process of cost minimization.

3.1.3 Statistical Mechanical Correspondence: Probability Distribution, Partition Function, and Observables

To clarify that the introduction of the Hamiltonian in this section is not just a metaphor, we specify the probability distribution of the social state. We set the probability of realizing state $\{\vec{\sigma}\}$ as an effective canonical distribution (in the sense of not tracking all non-equilibrium details):

$$P(\{\vec{\sigma}\}) = \frac{1}{Z}\exp\bigl(-\beta_{\text{eff}} H(\{\vec{\sigma}\})\bigr), \quad Z = \int \exp\bigl(-\beta_{\text{eff}} H\bigr)\, d\vec{\sigma} \tag{2}$$

[1]This normalization is a mathematical treatment to handle "direction of intent (Alignment)" and "magnitude of capability" separately. Differences in capabilities and heterogeneity of agents are considered in the generalized labor tensor $\vec{w}_a$ described later.

(where $\beta_{\text{eff}} := 1/T_{\text{eff}}$). Here, Z is the partition function, and macroscopic observables are defined as expected values regarding this distribution.

In this case, (i) J_{ij} is interpreted as a coupling expressing "cooperation/repulsion between agents" with a sign, (ii) $\vec{h}_i$ as an external field such as norms, protocols, and narratives, and (iii) μ as the market sensitivity of the entire interaction term (the strength of the impact of the system on price). Furthermore, the introduction of constraint Φ_k corresponds to the operation of excluding states that society does not tolerate from the support of the probability distribution, and $\lambda_{core} \to \infty$ mathematically implements "zeroing out the probability."

Economically, H is positioned as an approximation of the "social objective function behind observed prices, allocations, and behaviors (including externalities, frictions, and norm violations)." Therefore, policy and protocol design can be reinterpreted as a mechanism design problem of whether deforming the landscape of H through $\mu, \vec{h}_i, \lambda_k$ can simultaneously improve (a) stability (volatility suppression) and (b) well-being.

Moreover, if we introduce the average alignment as an example of an Order Parameter:

$$m := \left| \frac{1}{N} \sum_{i=1}^{N} \vec{\sigma}_i \right| \tag{3}$$

$m \approx 0$ represents high entropy (fragmentation/confusion), while $m \approx 1$ represents low entropy (consensus/cooperation). A "phase transition" in this paper is defined as a sudden change in m or an explosion in susceptibility (e.g., $\partial m/\partial \mu$).

Work volatility V is redefined as a function of "social temperature" T in this system. However, please note that the temperature here is not the temperature in equilibrium thermodynamics, but the Effective Temperature in a Non-Equilibrium Steady State (NESS).

$$V \propto T_{eff} = \left(\frac{\partial S}{\partial E} \right)^{-1} \cdot \frac{1}{\mathcal{L}_{info}} \tag{4}$$

Here, T_{eff} is the effective temperature and $\mathcal{L}_{info}$ is "Information Liquidity." This model reviews the equivalence with simple temperature and redefines volatility as a composite function of "information viscosity (market friction)" and "thermal fluctuations."

3.1.4 Operational Definitions: Estimating Volatility and Information Liquidity

To make the theory testable, this paper specifies V as an observable quantity. For example, using the log return $r_t := \log p_t - \log p_{t-1}$ of task price p_t, with a rolling window width W:

$$V_t := \sqrt{\frac{1}{W} \sum_{\tau=t-W+1}^{t} (r_\tau - \bar{r}_t)^2} \tag{5}$$

is set as the estimator for volatility ($\bar{r}_t$ is the same-window average). By comparing this V_t with the estimator for T_{eff} described later, we ensure the identifiability of the model. Practically, it can be connected to a framework that decomposes which factors are causing the variation in V_t through regression and state-space models using aggregate statistics of μ, J_{ij} and $\mathcal{L}_{info}$ as explanatory variables, thus estimating the effects of policy interventions (λ_{policy} and update frequency).

Next, to define $\mathcal{L}_{info}$ not by "order book depth" but by "semantic depth," we index the predictability between task description $d(T)$ (specs, minutes, code diffs, etc.) and execution log $\ell(T)$ (deliverables, evaluations, audit logs, etc.). Simply, using the conditional code length (cross-entropy) based on a language model:

$$\mathcal{L}_{info}(T) := \exp\big(-H_\theta(\ell(T) \mid d(T))\big) \tag{6}$$

is set (where H_θ is the negative log-likelihood by a model with parameter θ). Intuitively, the better the log can be "predicted" from the specs, the smaller the information friction and the larger the $\mathcal{L}_{info}$.

In this case, even with the same T_{eff}, if $\mathcal{L}_{info}$ is low, V can be amplified. Therefore, the main lever for policy intervention (protocol design) is not simply injecting liquidity, but raising $\mathcal{L}_{info}$ through (i) readability of specifications, (ii) auditability, and (iii) preparation of shared representations (ontology) for consensus building.

As noted in the critical analysis by AI, an increase in superficial liquidity (Quote Liquidity) in traditional HFT (High-Frequency Trading) can conversely trigger flash crashes. Therefore, $\mathcal{L}_{info}$ defined here is redefined not by order book depth but by "Semantic Depth"—that is, how deeply participants share the context of the task. This prevents the system from running away due to "apparent liquidity."

That is, large fluctuations in task prices (high volatility) are phenomena that can be observed near the critical point where system entropy S increases and the social structure undergoes a phase transition (order collapse or reconfiguration). While it is suggested that worker behavior might aim for gain acquisition at this stage, additional empirical evidence is needed to generalize this reality.

3.1.5 Estimation and Identification Protocol (Strengthening Identifiability)

To ensure that the parameter introduction in this paper is not arbitrary, we specify the estimation procedure (operational protocol) from observables. The aim is to fix (i) the set of estimable parameters, (ii) data requirements for identification, and (iii) handling of uncertainty in advance, so that the simulations in subsequent chapters are verifiable hypotheses rather than "selection of convenient values."

Estimation of Coupling J_{ij} (Interaction Network): Interaction J_{ij} is estimated from networks based on joint work, transactions, and mutual evaluation. Simply put, by measuring lagged correlations $C_{ij}(\Delta t)$ from the time series of state vectors, it can be defined as a regularized estimation:

$$\hat{J} = \arg\min_{J} \sum_{i \neq j} \left(C_{ij}(\Delta t) - f(J_{ij})\right)^2 + \gamma \|J\|_1 \tag{7}$$

(where f is a modeled response function and γ is the regularization for sparsity). What is important is that J_{ij} is estimated not as a "psychological or political evaluation" but as a **log-derived interdependence structure**.

Estimation of Market Sensitivity μ (Intervention Response): μ is estimated from the volatility response (or susceptibility) observed before and after protocol updates or incentive changes. For example, using intervention dummy $u_t \in \{0, 1\}$:

$$V_t = a + b\, u_t + c\, \widehat{\mathcal{L}}_{info,t} + \varepsilon_t, \tag{8}$$

b can be identified through a regression/state-space model (allowing time-varying coefficients) and connected to the operational estimator of μ.

Handling of Constraint Weight λ_k (Separation of Hard/Soft): $\lambda_{core} \to \infty$ is a mathematical expression, and in implementation, it is handled as **exclusion of states (filtering)**. That is, for candidate state S:

$$\Phi_{core}(S) = \text{False} \;\Rightarrow\; P(S) = 0 \tag{9}$$

is applied, and it is not included in the search/update rules. Meanwhile, λ_{policy} is a trade-off coefficient for policy purposes, and in implementation, it is a **design variable determined in the consensus process**, updateable only under (i) version control, (ii) fixation to audit logs, and (iii) public period for changes (consistent with freezes/rollbacks in Chapter 7).

Separation of T_{eff} and $\mathcal{L}_{info}$ (Avoiding Confusion): When using V as an observable, identification becomes unstable if the contributions of T_{eff} and $\mathcal{L}_{info}$ are confused. Therefore, this paper assumes a two-stage estimation where (i) $\mathcal{L}_{info}$ is independently estimated from the predictability of task specs and logs, and (ii) then the residuals of V are interpreted as variations in T_{eff}.

3.1.6 Sensitivity Analysis and Uncertainty Propagation (Presenting Parameter Rationale)

Since uncertainty always remains in estimated values, we do not fix conclusions with a single point estimate but also provide (i) prior distributions/estimation intervals, (ii) sensitivity analysis, and (iii) robustness of main conclusions. Specifically, for perturbations of parameter set $\theta := (\mu, J, \lambda, \cdots)$, local sensitivity of main indicator Y (e.g., constraint violation rate, ΔH, rollback frequency):

$$S_\theta := \frac{\partial Y}{\partial \theta} \tag{10}$$

and variance decomposition based on sampling (e.g., Sobol indices) explicitly state "which assumptions dominate the conclusion." As a result, subsequent stress tests can be reinterpreted not only as "searching for a dangerous future" but also as **experiments to identify parameters (uncertainty bottlenecks) that influence conclusions**.

3.2 Latent Torus: Negentropy Generation Mechanism

As system complexity increases, there is a risk of society becoming unstable, exceeding the limits of human cognition. Here, the "Latent Torus Concept" can function as a negentropy (order) generation mechanism that resists entropy increase (see Figure 5).

3.2.1 Landauer's Principle and Thermodynamic Cost

The generation of negentropy (order) in society by the Latent Torus involves a thermodynamic cost, as any erasure or processing of information in a physical substrate results in heat dissipation according to Landauer's principle (Landauer 1961; Bennett 1982).

$$\Delta Q_{exhaust} \geq k_B T \ln 2 \cdot \Delta I_{erased} \tag{11}$$

This thermal energy ΔQ is discharged outside the system as physical power consumption. Thus, the model functions as a "Social Heat Pump" that consumes physical energy to lower social entropy (informational disorder), ensuring long-term systemic stability.

3.3 Rigorous Definition of Latent Torus Concept: Information Asymmetry and Event Horizon

The "Latent Torus ($\mathcal{D}$)" proposed in this study is physically defined not as a mere security area but as an "Information Event Horizon" in the social system.

Definition 3.1 (LatentTorus $\mathcal{D}$)**.** In a social system $\mathcal{S}$, a subspace $\mathcal{D} \subset \mathcal{S}$ satisfying the following conditions is defined as a Latent Torus:

1. **Irreversibility of Information**: While information flow (transaction records) from the frontend $\mathcal{F}$ to $\mathcal{D}$ is possible, the flow of internal states (fine structure of the Hamiltonian) from $\mathcal{D}$ to $\mathcal{F}$ is information-theoretically prohibited. That is, the internal entropy $S(\mathcal{D})$ of $\mathcal{D}$ is unknowable to an external observer.

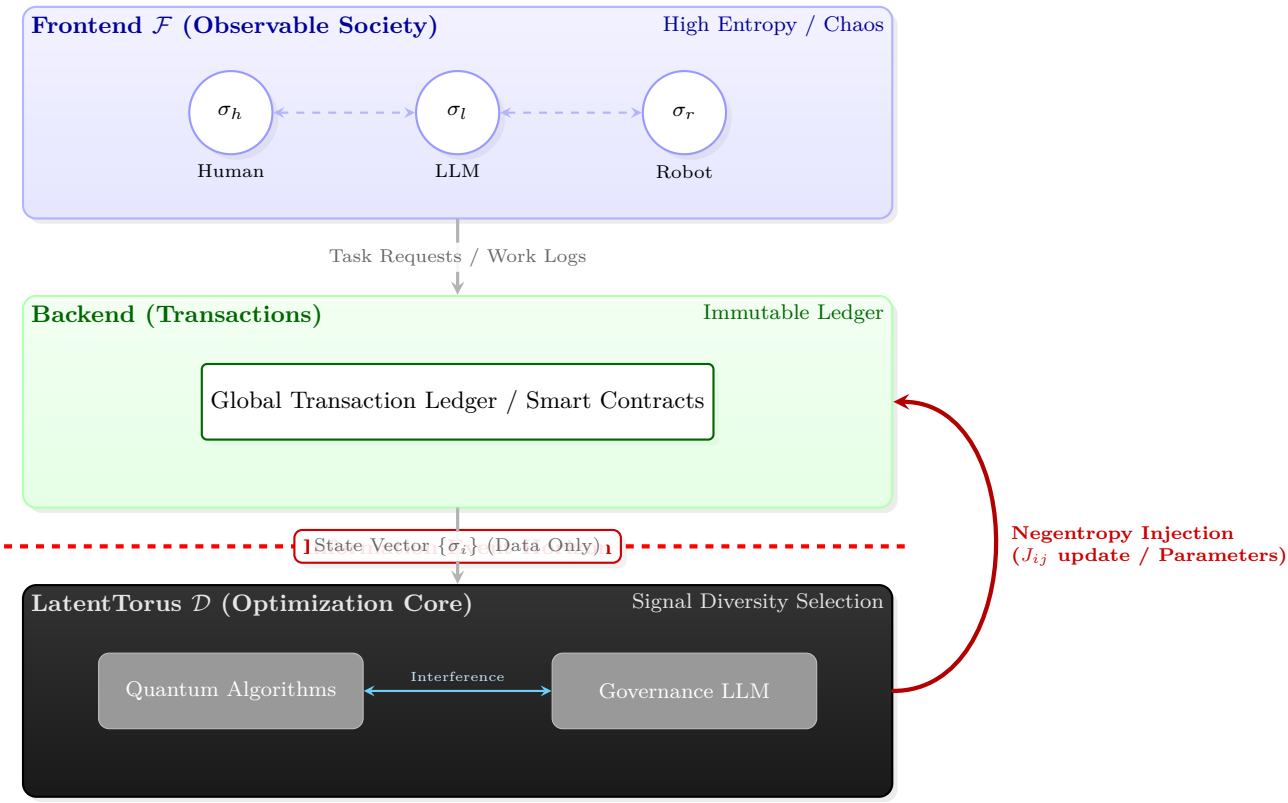

Figure 5: Three-layer structure in work volatility theory and the information event horizon. The Latent Torus absorbs social entropy and recirculates only optimized parameters (coupling constants) to the backend.

2. **Computational Completeness**: $\mathcal{D}$ contains all calculation resources and algorithms necessary to calculate the time evolution of the social system state vector $\{\sigma_i\}$.

3. **Mathematical Verifiability**: While the internal processes of $\mathcal{D}$ are a black box, it is mathematically proven through zero-knowledge proofs (zk-SNARKs) that its output complies with the Constitutional Core. This enables agents to verify legitimacy without knowing the content.

3.4 Verifiable Concealment: Architecture Using zk-SNARKs

To ensure both the integrity and mathematical verifiability of the "black-box" Latent Torus, this system adopts a "Verifiable Concealment" architecture. Using zk-SNARKs (Ben-Sasson et al. 2014; Groth 2016), we demonstrate that...

In this architecture, the statement to be verified is defined as the following logical conjunction:

$$\mathcal{R}(x, w) \iff (\Phi_{core}(S_{t+1}) = \text{True}) \wedge (H(S_{t+1}; w) \leq H(S_t; w)) \tag{12}$$

- **Constitutional Suitability** $\Phi_{core}(S_{t+1})$: The generated next-phase social state S_{t+1} does not infringe upon basic human rights or physical laws (Constitutional Core).

- **Pareto Improvement** $H(S_{t+1}) \leq H(S_t)$: The new state reduces (or keeps equal) the social cost compared to the current state. Maximizing the well-being term is also included here.

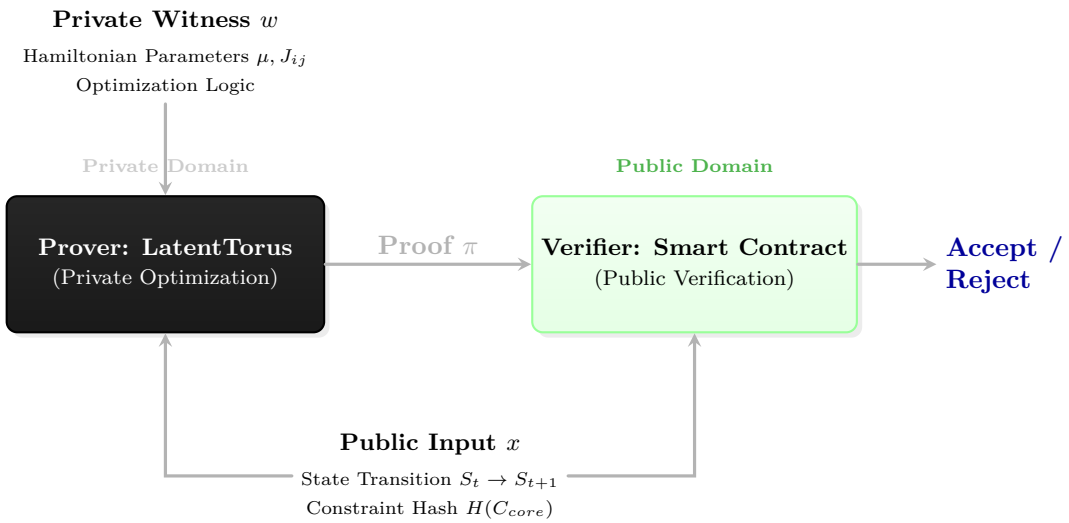

Figure 6: Verifiable concealment architecture using zk-SNARKs. The Prover (Latent Torus) submits only a proof π that the output satisfies constitutional constraints (Public Input) to the Verifier (Blockchain) without revealing private information (Witness).

The Latent Torus generates evidence π that satisfies this relationship $\mathcal{R}$:

$$\pi \leftarrow \text{Prove}(PK, x, w) \tag{13}$$

where PK is the proving key. Agents in the frontend verify this using the verification key VK.

$$\text{Verify}(VK, x, \pi) \in \{0, 1\} \tag{14}$$

Through this process, citizens can verify that the result "follows the constitution (protocol)" based on computational assumptions without knowing the details of the internal calculation process of the Latent Torus. This is a powerful design choice for balancing "Verify, don't trust" with optimization.

What is proven and what is not (Clarification): While the zk design in this paper makes it verifiable that output S_{t+1} satisfies constraints and does not worsen the objective function (and audit log consistency), the following information is **not included in the proof target**:

- Citizens' individual data (raw input values) and the internal search process (search history/intermediate states) of the Latent Torus.
- The substance of internal parameters w (entire J_{ij} matrix, model weights, private parts of prompts, etc.).
- "Global optimality" itself: What is proven is **constraint satisfaction** and **monotone improvement**, and optimality depends on the operational search strategy.

Therefore, system design focuses on (i) what is fixed as public input x, (ii) how much of the auditable log (hash chain) is included, and (iii) how to design freezes, rollbacks, and fallbacks during exceptions.

Based on this definition, we formalize the functions of the Latent Torus from a thermodynamic perspective. The entropy S_{total} of the entire social system normally increases due to the second law of thermodynamics

(increasing complexity and disorder in society).

$$\frac{dS_{total}}{dt} \geq 0 \tag{15}$$

However, the Latent Torus functions as Maxwell's demon (Leff and Rex 2003), maintaining the negentropy (order) of the frontend by discharging entropy from the system (selecting and erasing information).

$$\frac{dS_{\mathcal{F}}}{dt} = \frac{dS_{total}}{dt} - \frac{dS_{\mathcal{D}}}{dt} < 0 \tag{16}$$

where $\frac{dS_{\mathcal{D}}}{dt}$ is the amount of information (exhaust heat) absorbed and processed by the Latent Torus, and this processing capability determines the survival limit of the social system. To achieve this, sophisticated optimization algorithms working on quantum computers (Quantum Optimization Algorithms) rather than traditional silicon computers are essential inside $\mathcal{D}$.

Figure 7 shows the simulation results of this negentropy generation process. As the Latent Torus operates (Phase Shift), it can be confirmed that the disorder (entropy S) of the social system significantly decreases, and ordering progresses. In the figure, "deepend" is a tentative name for the Latent Torus.

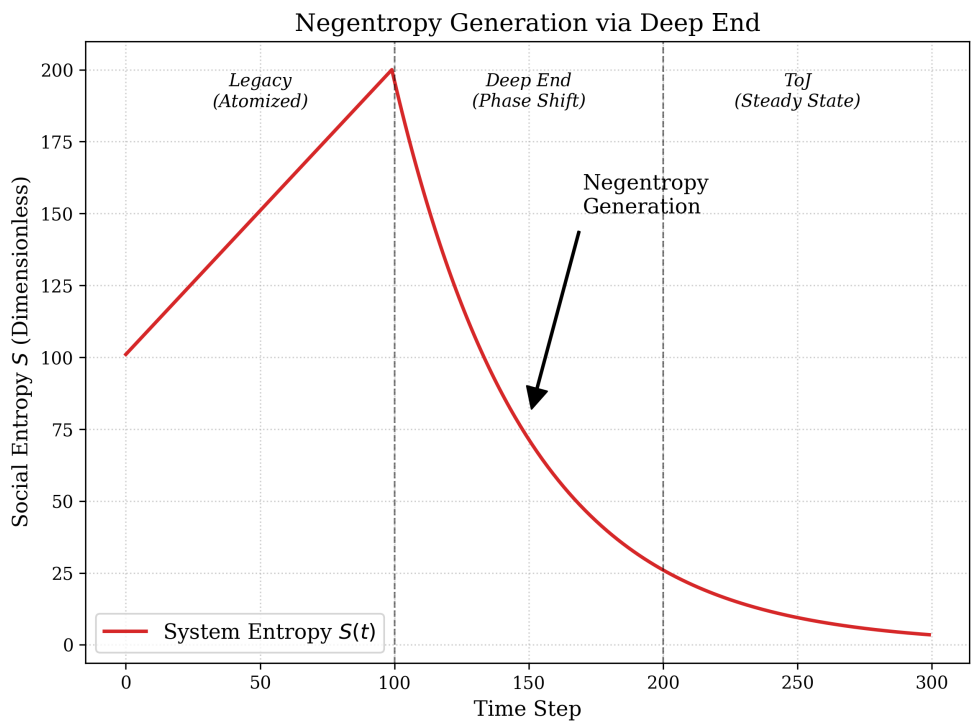

Figure 7: Simulation of negentropy generation by the Latent Torus. Through system intervention (Phase Shift), the social entropy that had been increasing turns into a decrease, shifting toward an ordered steady state.

Furthermore, from a geometric perspective (information geometry), the Latent Torus functions as a field that dynamically bends the metric tensor g_{ij} (Fisher information metric) on the social state manifold M.

$$g_{ij}(\theta) = \mathbb{E}\left[\frac{\partial \log p(x|\theta)}{\partial \theta_i}\frac{\partial \log p(x|\theta)}{\partial \theta_j}\right] \tag{17}$$

Semantic intervention by regulated LLMs is none other than manipulating this metric g_{ij} to shorten the "distance" of specific social transitions (e.g., transition to "environmentally friendly tasks"). The Latent Torus is a concept concerning the existence of a "singularity" that engineers the physical laws of society themselves.

3.5 Interference Model between Regulated LLMs and Quantum Computing

Inside the Latent Torus $\mathcal{D}$, a Governance LLM exists alongside the physical optimization by the quantum computer. The role of the Governance LLM is to translate cold mathematical optimization solutions into "semantics" acceptable to human society, and vice versa.

However, these two entities interfere with each other. Semantic intervention by the Governance LLM (rewriting J_{ij}) acts as a disturbance (noise) for the quantum system. This interaction is described by the following coupling model:

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}_{sys}, \hat{\rho}] + \sum_k \left(L_k \hat{\rho} L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \hat{\rho}\} \right) \tag{18}$$

where $\hat{\rho}$ is the density matrix of the social state and L_k is the Lindblad operator representing semantic intervention by the Governance LLM. The Governance LLM intervenes by modifying the coupling constants J_{ij}, a non-Hermitian operation that breaks unitarity. This is mathematically justified as the social system is modeled as an "Open Quantum System" interacting with an external semantic field; this dissipation term serves as the driver for state transitions away from unstable equilibria.

However, to maintain control-theoretic stability, it is necessary that the time scale t_{LLM} of semantic intervention by the LLM is sufficiently slow compared to the market relaxation time t_{relax} (adiabatic approximation condition).

$$t_{LLM} \gg t_{relax} \tag{19}$$

If $t_{LLM} \approx t_{relax}$, the system undergoes "Semantic Resonance" and leads to chaotic divergence. To prevent this, the update cycle of the Governance LLM is intentionally designed as "low-frequency."

Here, $\hat{V}_{LLM}(t)$ is the semantic potential term by the Governance LLM. Natural language text T is projected into physical variables of the Hamiltonian by the "Semantic Operator" $\mathcal{M}$.

Definition 3.2 (Semantic Operator $\mathcal{M}$)**.** A map that transforms natural language text (prompts or constitutional descriptions) Txt into energy bias terms of the Hamiltonian.

$$\mathcal{M} : Txt \to \mathbb{R}^N, \quad V_{LLM}(Txt) = \sum_i \alpha_i \cos(\theta_{i,target}) \tag{20}$$

where $\cos(\theta_{i,target})$ is the cosine similarity between the behavior vector of agent i and the "Constitutional Vector" contained in text Txt. Through this projection, ambiguous meaning is converted into a physical energy gradient.

3.5.1 Gauge Transformation for "Semantic" → "Physical" Conversion: Fixing the Reference Frame and Invariants

The conversion $Txt \to H$ in this paper is not a mere "embedding → scalar" projection but includes **fixing the reference frame (gauge)**. Gauge here corresponds to the redundancy of expression (degrees of freedom in coordinates) where the same normative content can take different expressions due to synonyms, paraphrasing, or contextual differences.

First, obtain potential vector $u \in \mathbb{R}^d$ from text Txt (embedding). Next, using matrix $E_{\mathcal{O}} \in \mathbb{R}^{d \times M}$ consisting of column vectors of institutionalized concept set (ontology) $\mathcal{O} = \{o_m\}_{m=1}^M$, map u to "conceptual coordinates."

$$c := G(u) := E_{\mathcal{O}}^\top u \in \mathbb{R}^M \tag{21}$$

In this case, even if u fluctuates in the neighborhood due to paraphrasing, the identity of the norm is maintained as long as the **coordinates** c **along the concept ID** ($\mathcal{O}$) are preserved. Furthermore, if the

difference in expression systems (model update/vocabulary rotation) is modeled as coordinate transformation $g \in \mathcal{G}$:

$$u' = g \cdot u, \quad E'_{\mathcal{O}} = g \cdot E_{\mathcal{O}} \tag{22}$$

(e.g., g as an orthogonal or affine transformation), then:

$$c' = (E'_{\mathcal{O}})^\top u' = (gE_{\mathcal{O}})^\top (gu) = E_{\mathcal{O}}^\top u = c \tag{23}$$

and c can be interpreted as a gauge invariant. That is, the **invariant part of the norm is fixed by conceptual coordinates c, and surface linguistic expression is allowed as a gauge**.

Finally, map c to the external field term:

$$\vec{h}(Txt) := W_h\, c, \qquad V_{LLM}(Txt) := -\sum_i \vec{h}(Txt) \cdot \vec{\sigma}_i \tag{24}$$

is defined (W_h is institutionally auditable linear map tied to hashes of prompts and constraint sets). Thus, natural language is connected to the Hamiltonian not as a "Constitutional Vector" but as an **invariant external field via conceptual coordinates c**.

3.5.2 Is Ambiguity "Heat" or "Potential Shape"? Dual Inflow Paths

Ambiguity in natural language can flow into the system in two ways: (A) as **effective temperature (heat)** from stochastic fluctuations, or (B) as **potential shape changes** of the objective function landscape. This paper separates the two and connects them to observables.

First, assume the Governance LLM forms a posterior distribution $q(z \mid Txt)$ (z as latent intent) regarding the intent of the text (application conditions, exceptions, etc.), and measure ambiguity as entropy:

$$\mathcal{A}(Txt) := H\big(q(z \mid Txt)\big) \tag{25}$$

(A) Incorporation as heat: When ambiguity is high, uncertainty in consensus building and auditing increases, and noise is injected into microscopic update rules. This can be incorporated as Langevin-type effective temperature:

$$T_{\text{eff}} \leftarrow T_{\text{eff}} + \kappa_T\, \mathcal{A}(Txt) \tag{26}$$

($\kappa_T > 0$).

(B) Incorporation as potential shape change: When ambiguity is high, the enforcement power of external field $\vec{h}$ is weakened (update width shortened) to suppress the risk of misguidance. This can be defined as regression based on uncertainty:

$$\vec{h}(Txt) \leftarrow \frac{1}{1 + \kappa_h\, \mathcal{A}(Txt)}\, \vec{h}(Txt) \tag{27}$$

The gradient of the potential itself by the external field relaxes (the landscape flattens).

Therefore, while ambiguity can act as both "heat" and "landscape deformation," it is unified operationally by a control rule: **at high ambiguity, reduce enforcement power (external field) toward a freeze and transfer to fallback if necessary** (consistently with stress tests in Chapter 7).

3.5.3 Semantic Hacking and Safety Requirements

Since the Governance LLM intervenes in society, the attack surface extends not only to "code" but also to "meaning." Semantic hacking here refers to attempts to guide the system in unintended directions through (i) contamination of input narratives, (ii) Goodhartization of proxy indicators (evaluation function hacking), and (iii) constraint deviations accompanying distribution drift.

In this study, the following three-layer defense is imposed as a safety requirement:

1. **Specification Layer**: Implement Φ_{core} as a formal constraint, and exclude red lines such as anti-discrimination, due process, and privacy from the state space.
2. **Learning Layer**: Detect early the correlation collapse between objective function and reality through anomaly detection, distribution drift detection, and counterfactual tests.
3. **Operational Layer**: Enable stopping irreversible runaway through auditing (log and hash chain), red teaming, update freezing, and Democratic Veto (Human Veto).

As an empirical design, stress tests are mandatory where the change in H and the constraint violation rate when giving adversarial inputs (polluted prompts or biased data) are compared with and without intervention.

Specifically, this means managing "Human Leap." Traditional capitalism depended on growth through disordered innovation (leap), but that simultaneously caused system instability. The Governance LLM keeps this leap within a controllable range through semantic intervention and performs a kind of "life-support operation" on the recursive processes of the Latent Torus and the frontend. This makes it possible to prevent the collapse of the entire system while continuing calculable growth even under population decline or low growth.

3.6 Open Market Operations by Smart Contracts

Market intervention using LLMs is not merely supply-demand adjustment. it is a "semantic intervention" that dynamically rewrites J_{ij} (coupling constants). If the value of a specific task crashes (volatility becomes excessive), the system redefines the narrative (social significance) related to that task via smart contracts and forcibly manipulates demand h_i. This is an "open market operation" of values themselves, exceeding interest rate manipulation by central banks, and will be a major governance technology in post-capitalism.

3.7 Unified Description of Heterogeneous Agents: Generalized Labor Tensor and Ontological Arbitrage

The core of the "Transaction of Japan" concept is processing labor subjects with fundamentally different physical foundations—humans, LLMs, and humanoids (physical robots)—on a unified mathematical model. This section defines the "Generalized Labor Tensor" that enables this.

3.7.1 State Vectorization of Agents

All agents $a \in \mathcal{A}$ ($\mathcal{A} = \mathcal{A}_{Human} \cup \mathcal{A}_{LLM} \cup \mathcal{A}_{Robot}$) are described as a state vector $\vec{w}$ with three components:

$$\vec{w}_a = \begin{pmatrix} E_a \\ I_a \\ \Delta S_a \end{pmatrix} \tag{28}$$

3.7.2 Normalization and Dimensionless Quantities

To handle agents with different physical foundations on the same Hamiltonian, normalization of each physical quantity is essential. This study introduces the following "Human-Standard" Characteristic Scales:

- $E_0 \approx 10^7$ J (Human Daily Metadata): The amount of energy corresponding to the basal metabolism of a human per day.

- $I_0 \approx 10^2$ bps (Cognitive Limit): The limit of human linguistic information processing capability.
- S_0: The standard deviation of the degree of freedom in an individual's behavioral choices (or an economic constant corresponding to the Boltzmann constant k_B).

From a physical perspective, these constants should be interpreted as "Renormalization Constants" in Effective Field Theory. Since it is impossible to directly describe macroscopic social phenomena from microscopic physical constants $(\hbar, k_B)$, physical quantities are redefined (Coarse-graining) at the intermediate scale of humans.

Using these, each component of the generalized labor tensor $\vec{w}_a$ is converted into dimensionless quantities $(\hat{e}, \hat{i}, \hat{s})$:

$$\vec{w}_a = \begin{pmatrix} \hat{e}_a \\ \hat{i}_a \\ \hat{s}_a \end{pmatrix} = \begin{pmatrix} \eta_{bio-elec} \cdot E_a/E_0 \\ I_a/I_0 \\ \Delta S_a/S_0 \end{pmatrix} \tag{29}$$

Here, $\eta_{bio-elec}$ is the "Bio-Electric conversion efficiency coefficient." This is a parameter for correcting differences in economic value in the market between biological energy (Joule) and electric energy (Wh), determined dynamically from the relative prices of actual "labor cost" and "electricity bill."

This allows, for example, comparing and operating LLMs as "existences with small $\hat{e}$ but extremely large $\hat{i}$ (10^6 or more)" and humans as "existences with small $\hat{i}$ but large $\hat{s}$ (creative fluctuations)" on a unified vector space. State vector $\vec{\sigma}_a$ in the Hamiltonian is a further normalization of this dimensionless $\vec{w}_a$ ($\vec{\sigma}_a = \vec{w}_a/|\vec{w}_a|$).

3.7.3 Equivalent Exchange of Labor and "Ontological Arbitrage"

In the "Latent Torus" on smart contracts, task T is also defined as a vector of required $(E, I, \Delta S)$. The system searches for the optimal agent combination (linear combination) to satisfy this task vector.

$$\vec{T} \approx \sum_i c_i \vec{w}_{a_i} \quad (c_i \geq 0) \tag{30}$$

Here, coefficient c_i indicates the contribution of each agent. What this equation means to predict is that in tasks where "creativity ΔS is not required and only information processing I is sought," humans (high cost, slow speed) will likely be fully compatible with LLMs (low cost, high speed).

Conversely, in situations where high ΔS (creative jumps) is required, the value of humans is "reevaluated" as it cannot be replaced by the stochastic outputs of LLMs. We call the process of optimizing using these differences in cost and capability among heterogeneous agents "Ontological Arbitrage." Thus, the labor market transforms from a place to sell "human time" into a physical field for the "optimal placement of physical and informational resources."

4 Architecture of Legal Engineering

It should be emphasized here that the term *Legal Engineering* itself has been used in previous research and practical contexts (Goldenfein and Leiter 2018; De Filippi and Hassan 2016; Governatori et al. 2018; Soavi et al. 2022; Rahwan 2018; Surden 2012); this paper does not claim to have coined the term. The novelty of this paper lies in placing **algorithm-based law implementation** at its core and explicitly defining an integrated design principle that (i) embeds the Constitutional Core into the state space as hard constraints,

(ii) operates the policy layer as constrained optimization, and (iii) guarantees legitimacy through zk-based verifiability.

When this system is implemented in society, how should the existing legal system be aligned and integrated with algorithms (Code)? In this chapter, based on the model construction in the previous chapters, we define the architecture of "Legal Engineering," which redefines law from "clauses to be interpreted" to "formal specifications to be implemented." If traditional jurisprudence has remained in "software operation," discussing the interpretation and application of law, Legal Engineering takes on the "description of the kernel," designing and building the architecture of the social system itself.

4.1 Algorithmization of Law

4.1.1 Triple-Layer Structure of Law and Recursive Structural Consistency

Law in a Latent Torus society is strictly structured into the following three layers according to its changeability and abstraction. This hierarchical structure ensures the "Reference" for verification by zk-SNARKs.

- **Layer 0: Meta-Constitution (Fixed Point)**: The physical and logical foundation of the system. This is a system of axioms defining the laws of thermodynamics, cryptographic hardness, and the "inviolability of basic human rights." It functions as a prerequisite that neither humans nor algorithms can directly change, serving as the reference point for all verifications.
- **Layer 1: Constitutional Core (Variable Boundary)**: The basic law based on the social contract. This includes the scope of application of specific human rights and the ethical consensus at that point in time. It can be modified (Amendments) over a long period through the accumulation of "dispute terms." However, the amendment itself must be mathematically proven to comply with Layer 0.
- **Layer 2: Algorithmic Civil Law (Fluid Civil Law)**: A set of parameters that fluctuate instantaneously (interest rates, tax rates, traffic regulations). These are optimized in real-time within the boundary conditions of Layer 1 to minimize the Hamiltonian H.

What is important here is the concept of "Recursive Verification." Every day, the Latent Torus proves that the parameter optimization results of Layer 2 are consistent with the current Layer 1 (Proof daily). On the other hand, when Layer 1 itself is updated (the constitution is amended), the system must generate a "Meta-Proof update" proving that "the new Layer 1 complies with the immobile Layer 0 and reduces the expected dispute energy of society as a whole more than the old Layer 1." Through this recursive structure, we aim to provide at least a formal reference point for the circular reference problem: "If the constitution itself is rewritten, what should we believe in?"

4.1.2 Solution to the Initial Value Problem: Refactoring, Not Revolution

Here, the bootstrap problem arises: "Who decides the first Layer 1 (Genesis Constitution)?" This system starts with a "snapshot of existing laws." That is, on the first day of system operation (Day 0), the current Constitution of Japan and the Civil Code are translated into a Formal Language and set as the provisional initial values. Therefore, transition to the Latent Torus is positioned not as a "hard fork (revolution)" that destroys the existing order, but as a "Legacy Migration" process that incrementally rewrites legacy and inefficient code (natural language laws) into formal specifications with high verifiability. This is expected to improve operational efficiency while maintaining legal continuity.

4.1.3 Roadmap for the Transition Process: Legal Compiler

The process of translating natural language law (legacy) into formal specifications is called a "Legal Compiler," organized as a phased transition:

1. **Shadow Spec**: Formalize current laws and run them in parallel, maintaining actual operations as before, while verifying the consistency (contradiction detection, boundary conditions) of the formal specifications.
2. **Automated Execution in Limited Domains**: Introduce automated execution based on formal specifications starting from areas with relatively low risk and high auditability, such as traffic, subsidies, and contracts.
3. **Case-law Bridge**: Reflect analog judgments (human judgments) as inputs for specification updates, establishing a circuit where case law is "compiled" into formal specifications.

Handling of Interpretation Fluctuations (Specification Uncertainty): Since unavoidable ambiguities remain in the translation to formal specifications, "specification uncertainty" must be treated as a design target during the transition period. For example, by providing an ensemble of multiple specifications (interpretation candidates) or confidence intervals, and branching to "specification update procedures" instead of optimization when a point of contention arises, exception handling functions as a safety valve.

4.1.4 Mathematical Proof: Ontological Stability via Recursive zk-SNARKs

To describe the "recursive structural consistency" mentioned in the previous section in a computationally rigorous form, we formulate it using the IVC (Incrementally Verifiable Computation) model. Let the state of society as a whole at time t be $\mathcal{S}_t = (S_t, C_t)$, where S_t is the economic state vector (Layer 2) and C_t is the constitutional code (Layer 1) at that point. The proposition R for proving that the system has legitimately transitioned from $\mathcal{S}_t$ to $\mathcal{S}_{t+1}$ is defined as the logical conjunction of the following two partial verification functions:

$$R(\mathcal{S}_t, \mathcal{S}_{t+1}, \pi) = V_{state}(S_t, S_{t+1}, C_t) \wedge V_{const}(C_t, C_{t+1}, C_{meta}) \tag{31}$$

- V_{state}: A function verifying that the transition (Hamiltonian minimization) from state S_t to S_{t+1} was properly performed under constitution C_t.
- V_{const}: A function verifying that when the constitution itself is updated from C_t to C_{t+1}, the change is constrained by the fixed point C_{meta} (Layer 0) and has gone through proper procedures (reduction of dispute energy).

Recursive proof π_{t+1} is inductively generated with the proof of the previous time π_t as input:

$$\pi_{t+1} \leftarrow \text{Prove}_{IVC}(pk, (\mathcal{S}_{t+1}, C_{meta}), (\mathcal{S}_t, \pi_t)) \tag{32}$$

Here, Prove_{IVC} executes the verification circuit $\text{Verify}(vk, \mathcal{S}_t, \pi_t)$ internally, confirms that "all past history is legitimate," and then compresses the proof by adding the legitimacy R of the current step. Through this inductive structure, at any time T, the design aims to concisely confirm consistency from the start of system operation (Genesis) to the present simply by verifying a single proof Π_T (as a verification procedure) based on computational complexity.

$$\forall T > 0, \text{Verify}(vk, \mathcal{S}_T, \Pi_T) = 1 \implies \bigwedge_{t=0}^{T-1} R(\mathcal{S}_t, \mathcal{S}_{t+1}, \pi) = \text{True} \tag{33}$$

This represents the design intent to bring the concept of "law" closer from human-interpretable text to a mathematically verifiable "chain of computation."

4.1.5 Trade-off Between Legal Stability and Dynamic Optimization

A fundamental concern in dynamic legal systems is the potential loss of predictability for citizens. To address this, we propose a "separation of domains" strategy. Basic domains related to the right to life and physical freedom (Constitutional Core) should maintain high stability. On the other hand, in economic and administrative domains such as interest rates, tax rates, and traffic regulations (Algorithmic Civil Law), fixation can hinder adaptation to changes in the external environment and amplify practical uncertainty through inefficiency (legal lag).

Therefore, in this paper, we aim to balance optimization and predictability through the following design principles: (i) updating Layer 1 at low frequency with high cost (requiring consensus building and verification), and (ii) updating Layer 2 at high frequency with low cost while pre-constraining update rules (cycle, maximum change width, emergency stop conditions).

4.1.6 Consistency with Existing Laws and API

The connection with existing positive law functions as an API (Application Programming Interface). "Prohibited matters (theft, injury, etc.)" defined by current laws are incorporated into the Hamiltonian as "penalty terms (barriers of infinite cost)" in the agents' action space. What is important is the automation of Enforcement. Through smart contracts, contract non-performance and infringing acts become physically and economically difficult to execute (or compensation is immediately executed), reducing the need for post-hoc "dispute resolution" by courts. The role of law partially shifts from "resolving disputes" to "designing parameter spaces where disputes do not occur."

However, economically, if predictability and credibility of enforcement are lost, transactions may shrink, and the expected returns on investment and innovation may decrease. Therefore, "Rule of Code" should be understood as institutional design including (i) transparency of amendment procedures (Layer 1 update conditions), (ii) advance notification of impacts (Layer 2 update cycle), and (iii) clarification of the scope for exception handling (discretion).

4.1.7 Automated Debugging System: Meta-Legal Debugger

However, parameter adjustment by Algorithmic Civil Law alone may not be enough to fully prevent unknown vulnerabilities or unexpected conflicts between laws (deadlocks). Therefore, this system implements a high-level function "Meta-Legal Debugger" to monitor the consistency of law.

- **Constant Audit of Logical Consistency**: Constantly monitor whether logical contradictions have arisen between the Constitutional Core and Civil Law using Formal Verification. If the result of parameter optimization leads to a solution that infringes upon human rights provisions (Core), the Debugger immediately rolls back that transaction and corrects the parameter generation function itself.
- **Generation and Learning of Virtual Precedents**: Generate a large number of marginal dispute cases that have not occurred in reality on a simulation (Fuzzing) and test whether the current legal al-

gorithm can output appropriate solutions. This allows law to learn "future precedents" before problems occur and fix bugs preemptively (Pre-crime approach).

The "power of judicial review" traditionally held by the Supreme Court is automated by this Meta-Legal Debugger, and the legal system gains the ability to evolve autonomously (Self-Healing) without human intervention.

4.1.8 Functional Replacement of Licensed Professionals: Automated Minimization of Dispute Energy

In a traditional legal system, prosecutors and lawyers have played the role of aligning parameters such as a defendant's "guilt/innocence" or "sentencing" with social norms through linguistic argumentation. In physical terms, this is an extremely high-cost "adiabatic process" of finding an equilibrium point through the synthesis of opposing vectors. In a Latent Torus society, this process is formulated as a minimization problem of the "dispute energy term $H_{dispute}$" on the Hamiltonian and solved instantaneously by an algorithm.

$$H_{dispute} = \sum_{\langle i,j \rangle} J_{legal}(c_i - c_j)^2 + \sum_i \lambda_{norm}(c_i - c_{truth})^2 \tag{34}$$

Here, c_i, c_j are the parties' claim vectors, and c_{truth} is the objective truth vector recorded by the monitoring system. Crucially, we distinguish between physical truth and semantic context; while "A hit B" is a physical fact, "it was self-defense" is a legal interpretation. This system maintains this separation...

- **System 1: Physical Truth (c_{truth})**: Objective degrees of freedom determined with $P > 99.9\%$ by sensors, cameras, and multi-sig. Fixing only the fact of "hitting." Handled by the physical layer.
- **System 2: Semantic Context ($c_{context}$)**: Judgments of probability of "self-defense" where the LLM analyzes behavioral context, motives, and social background and maps the situation to a tensor space. Handled by the semantic layer.

The Hamiltonian $H_{dispute}$ is constructed to minimize the distance from the composite vector of these two, $\Phi = c_{truth} \otimes c_{context}$.

$$H_{dispute} = \sum_{\langle i,j \rangle} J_{legal}(c_i - c_j)^2 + \alpha_{phys}(c_i - c_{truth})^2 + \beta_{sem}(c_i - c_{context})^2 \tag{35}$$

This makes it possible to mathematically describe highly sophisticated legal disputes such as "admitting the fact but with a different motive." Furthermore, by having the LLM automatically generate these J_{legal} and weights α, β from past precedents (Generative Hamiltonian), the problem of defining costs for the Hamiltonian is also solved. The traditional dispute resolution process by licensed professionals can also be seen as solving the dispute terms extracted from society by an algorithm.

Quantitative Consideration (Disappearance of Licensed Professional Functions): Traditionally, minimizing this $H_{dispute}$ required iterative calculation by humans through evidence collection, trial, and deliberation, and its computational complexity (social cost) increased from $O(N^2)$ to $O(N^3)$ relative to the number of participants N. Meanwhile, in this system, since all events are recorded in the trust chain, c_{truth} is known. However, even with the introduction of quantum optimization, the computational complexity is generally not made constant; it **depends on the problem size, number of constraints, and hardware configuration**. In this paper, instead of an asymptotic $O(1)$ claim, we adopt an implementation evaluation that reports $p95/p99$ latency and solution quality for each instance class.

4.2 Redefining Money: The Fusion of Law and Money

Finally, we discuss the future of "Money." In the Latent Torus society proposed by this research, existing "fiat currency" may entrust part of its role to protocolized value transfer and verifiable credit, leading to a decrease in its relative importance. This is because fiat, which depends on the compulsory power of state authority, will increasingly be compared with systems based on cryptographic technology and smart contracts in terms of programmability and verifiability. In this system, the concept previously called "money" is decomposed into the following three physical and informational functions and reintegrated into the Grand Unified Algorithm.

1. **Unit of Account Function → Physical Energy and Hamiltonian (H)**: Traditional "prices" are strictly physically defined as energy terms (Joules) and information processing amounts (bps) in the Hamiltonian. Thus, economic value is described not as "market whims" but as "thermodynamic costs," and phenomena such as inflation and deflation become physically controllable as fluctuations in the system's temperature or volume.
2. **Store of Value Function → Social Credit Score (Trust Potential τ)**: The act of "saving" shifts from increasing the balance of a bank account to accumulating a credit score τ based on the history of contribution to society. This τ is saved as an immutable record on the blockchain (Immutable Ledger), more robust than non-rusting Gold.
3. **Medium of Exchange Function → Law as Smart Contracts (Code)**: Settlement is no longer the movement of coins from A to B. It is the "state transition" itself, where the optimization algorithm matches supply and demand (state vectors) and minimizes the Hamiltonian. This transition is defined by "Law," which simultaneously functions as "Currency."

That is, in the Latent Torus, "Code is Money." Parameter adjustments of law through Legal Engineering (equivalent to changing interest rates or tax rates) function directly as adjustments of money supply and wealth redistribution. The era when law and economy were separate has ended, and both are completely fused into a single code system called "Social OS."

5 System and Algorithms

In this chapter, we propose specific algorithms that operate on the Legal Engineering architecture defined in the previous chapter, along with system details to make it function as social infrastructure.

5.1 Extension from Satoshi Nakamoto: Proof of Trust (PoT)

The Bitcoin paper by Satoshi Nakamoto (Nakamoto 2008) solved "double-spending" on P2P networks through Proof of Work (PoW), realizing trustless consensus building. Furthermore, it has been shown that blockchain can be effective even in complex multi-attribute evaluations such as supply chains (Haffar and Özceylan 2024).

However, the "social consensus" handled by this study is not limited to confirming the legitimacy of payments. It is more multi-dimensional consensus building that includes elements such as individual credit, the social value of tasks, and community norms.

This problem also connects with trust evaluation in OSN (calculating the trust value of each node and connecting it to communication judgments). For example, a framework has been proposed where a central administrator judges the possibility of communication in real-time by combining a mathematical model of trust and decision-making (TOPSIS) (Rathee et al. 2021). The PoT proposed in this paper is characterized by reinterpreting this not as **centralized adjudication** but as the **weight of consensus building itself (voting power)**, and adding verifiability through zk.

Moreover, it has been pointed out in Hurwicz's mechanism design theory (Hurwicz 1973) and "Radical Markets" by Posner and Weyl (Posner and Weyl 2018) that price mechanisms (monetary incentives) alone have limits in the supply of public goods and the allocation of complex social values. To safely operate proposals such as Quadratic Voting or Common Ownership Self-Assessed Tax (COST), "non-transferable credit" linked to individual identity is required while deterring Sybil attacks.

Here, to supplement the intuition of trust formation, we use the "bank account balance" as a metaphor. Bank deposits function as "one's own money" not only because of the cryptographic representation of ownership (account name) but also because institutional procedures such as deposits, withdrawals, settlements, dispute claims (chargebacks), audits, and compensation are piled up. Similarly, CSCT will operate in limited domains starting from identity verification (KYC), accounting, and dispute processing by existing trust entities (banks, local governments, corporations, etc.) in the initial stage, and will incrementally reduce the "range of trust" through the disclosure of operation logs, third-party audits, dispute claims/rollbacks, and compensation schemes. Finally, by expanding verifiability through ZK, etc., the part depending on credit will be replaced by verification, and the Trust Bottleneck will be progressively mitigated.

This design is close to the concept of Soulbound Tokens (SBT) (Buterin, Weyl, and Ohlhaver 2022) proposed by Vitalik Buterin and others. The Trust Potential τ in this research is a redefinition of SBT as a dynamic physical quantity, and the system guarantees the accumulation of good deeds that cannot be purchased with money.

Furthermore, as the threat of quantum computers becomes real, traditional signatures depending on elliptic curve cryptography become fragile. Therefore, in this research, we propose "Proof of Trust (PoT)" as a new consensus algorithm using quantum-resistant cryptography (signatures based on the shortest vector problem of lattices, etc.).

5.1.1 Mathematical Model of Trust Potential

The social credit (Trust Potential) $\tau_i(t)$ of each agent i is dynamically updated by the past transaction history and the evaluation network from others.

$$\tau_i(t+1) = \alpha \sum_{j \in \mathcal{N}_i} \omega_{ji} \tau_j(t) \cdot \mathrm{sign}(T_{ji}) + \beta \mathcal{H}(i) \tag{36}$$

where α, β are weight coefficients, $\mathcal{N}_i$ is the set of adjacent nodes that transacted with agent i, ω_{ji} is the transaction amount (energy amount), and T_{ji} is the feedback (± 1) after the transaction. $\mathcal{H}(i)$ is a history function, which extends the concept of the "longest chain" in Nakamoto consensus to the "best evaluation chain."

This τ_i functions as Voting Power within the Latent Torus. That is, consensus building is weighted not by pure computational power (Hash Power) but by accumulated social credit.

5.1.2 Preventing the Matthew Effect: Decay and Redistribution of Trust

A concern for the PoT system is the "Matthew Effect (the rich get richer)," where agents who obtain high credit early continue to be advantageous. To prevent this, this system introduces the following "Trust Decay" mechanism:

$$\tau_i(t + \Delta t) = \tau_i(t)\, e^{-\lambda \Delta t}, \quad \Delta t > 0 \tag{37}$$

where λ is a decay constant. This paper adopts discrete-time updates (defaulting to $\Delta t = 1$). Since credit naturally decays over time like physical entropy, agents must constantly keep creating new value (tasks) to

maintain their status. This functions as an "inheritance tax-like reset mechanism" that prevents the fixation of hereditary interests.

5.1.3 Graph-theoretic Robustness: Resistance to Sybil Attacks and Collusion

The resilience of Proof of Trust (PoT) against Sybil attacks and collusion rings is a critical design requirement (Douceur 2002). In this system, we... The evaluation function (38) of this system has the property of "Eigenvector Centrality" in graph theory, rather than simple cumulative addition.

$$\vec{\tau} = \alpha W \vec{\tau} + \beta \vec{H} \tag{38}$$

where W is the evaluation matrix. As Google's PageRank algorithm shows, even if low-quality nodes form a "spam farm" where they highly evaluate each other, the credit score within that loop converges to a minimal value in the entire system unless a link (evaluation) is obtained from an external high-trust node (a hub conducting legitimate economic activities). Therefore, for an attacker to deceive the system, they must actually provide socially useful value and gain evaluations from existing high-trust agents. As a result, the cost of attack rises, and behavior close to legitimate activities may be required. Incentives for attacks are expected to be curtailed.

5.1.4 Emergency Stop Authority: Decentralized Supreme Court and Democratic Veto (Human Veto)

As a safety device against algorithm runaway (unexpected optimization of the objective function), the system has a "Decentralized Supreme Court." It is composed of a group of virtuous human agents (jurors) selected at random, who can exercise a "Democratic Veto / Human Veto" against the algorithm's decisions. This right is not used for active law generation, but only as an "emergency stop button" when the system is about to cross human rights red lines. To avoid hindering the immediacy of High-Frequency Trading (HFT), this veto is exercised only for the "Low-frequency Domain," such as updates of the Constitutional Core or setting parameter ranges for Algorithmic Civil Law, rather than individual transactions.

Minimum Protocol Specification (Proposal): To ensure Democratic Veto (Human Veto) does not end with a slogan, minimal specifications are defined as follows:

- **Target**: Of state transition proposals $S_t \to S_{t+1}$, updates of Layer 1 (Constitutional Core) or updates of the allowable range of Layer 2 (policy parameters).
- **Trigger Condition**: (i) When approaching the constraint boundary (e.g., the fulfillment margin of Φ_{core} is below the threshold), (ii) when audit indicators show anomalies (drift, inconsistency).
- **Period**: Within Δt from proposal disclosure (consistent with the update cycle of the low-frequency domain).
- **Threshold**: Established by PoT-weighted voting or signatures by a certain percentage of jurors.
- **Effect**: (a) Remand (re-optimization), (b) rollback, (c) transfer to analog fallback (human trial).

Selective Disclosure Protocol for Analog Fallback (Legal-Technical Draft): To enable effective Human Veto while maintaining "verifiable concealment," the disclosure of encrypted state vectors is implemented as **limited-purpose selective disclosure** rather than full disclosure. The minimum requirements are:

1. **Legislated Disclosure Triggers**: Disclosure occurs only when (i) a Veto is sustained, (ii) warrant-equivalent approval from judicial/audit bodies is obtained, and (iii) it is restricted to time-limited purposes (trial, audit, victim relief).
2. **Cryptographic Implementation**: State vectors x_t are held as per-attribute commitments (Merkle trees or Vector Commitments). Only required attributes are presented using ZK-range proofs or membership proofs. The original raw data remains hidden, with verifiers only confirming the "truth of the claim."
3. **Key Management and Privilege Separation**: Decryption keys are managed via threshold secret sharing (k-of-n) distributed across multiple entities (courts, auditing firms, citizen representatives). Single-entity disclosure is prohibited, and all disclosure operations are recorded in irreversible audit logs.
4. **Procedural Guarantees**: Subjects under review are pre-notified of the disclosure range, purpose, retention period, and appeal methods. Upon completion of proceedings, data is "re-sealed" in a manner that reduces re-identifiability.

This specification allows Human Veto to be defined as a system interface that satisfies **minimal disclosure, prohibition of purpose-outside use, and post-auditability**, rather than arbitrary intervention into a black box.

The division of roles can be found in the analogy of the cognitive process proposed by Daniel Kahneman:

- **System 1 (AI/LatentTorus)**: Fast, automatic, intuitive (statistical). Responsible for nanosecond-level HFT and daily dispute resolution (parameter adjustment). The "circuit breaker" here functions as a conditioned reflex-like autonomic nervous system against systemic risk.
- **System 2 (Human/Supreme Court)**: Slow, intentional, logical (ethical). Responsible for exception events that System 1 cannot handle and judgments requiring deliberation, such as constitutional changes.

By hard-coding two thinking systems with different speeds into the social OS, the framework achieves both efficiency and safety, effectively addressing potential time-scale mismatches in bubble control (Sornette 2003):

- **Anti-Singularity Control (System 1)**: For finite-time singularities ($t \to t_c$) such as bubble collapses, the "emotional circuit breaker ((41))" in the physical layer responds at ultra-high speed (millisecond order) $t_{breaker} \ll t_{market}$, preventing divergence by adding physical damping. No semantic interpretation intervenes here.
- **Anti-Uncertainty Control (System 2)**: Semantic intervention by the Latent Torus is executed on an adiabatic schedule $t_{LLM} \gg t_{relax}$ after time is gained by the above damping or at the omen detection stage.

That is, physical runaway is suppressed by "physical laws (code)," and subsequent structural defects are corrected by "meaning (AI)." Through this division of roles, we aim to reduce blind spots in control.

5.1.5 Restoration of Humanity: Introduction of the Well-being Term

As a departure from the "model using humans as an energy source" being criticized, we explicitly introduce the "Human Well-being" term W_{human} into the objective function of the Hamiltonian.

$$H_{total} = H_{sys} - \lambda W_{human} \tag{39}$$

The system bears the obligation to search for a Pareto optimal solution that maximizes human well-being W_{human} while minimizing the total cost of society H_{sys}. This is the implementation of "human rights protection" at the level of physical laws.

5.2 Social Consensus Building Algorithm and Crowd Psychology

Decision-making for society as a whole (e.g., changing the reward rate for a certain task category) is represented not as the sum of individual will vectors $\vec{v}_i$, but as a non-linear composite vector $\vec{\Omega}$ including interactions.

$$\vec{\Omega} = \tanh\left(\sum_i \tau_i \vec{v}_i + \sum_{<i,j>} K_{ij}(\vec{v}_i \cdot \vec{v}_j)\right) \tag{40}$$

The second term is an interaction term representing Crowd Psychology. K_{ij} is a "resonance coefficient" between agents, corresponding to the propagation of emotions through SNS, etc. According to the findings of behavioral economics (prospect theory), humans react excessively to losses (loss aversion). Therefore, during bubble collapses or panics, K_{ij} can increase rapidly, and there is a risk of the entire system sticking to extreme values (± 1).

To prevent this, the proposed algorithm incorporates an "emotional circuit breaker." This is a function that automatically dampens (cools) K_{ij} when the total amount of the crowd psychology term exceeds a threshold.

$$K_{ij}^{corrected} = \frac{K_{ij}}{1 + \gamma|\vec{\Omega}|^2} \tag{41}$$

This damping term γ is not a constant, but an adaptive function $\gamma(\mathcal{T})$ based on network topology analysis. A primary design consideration is whether uniform damping might inadvertently inhibit legitimate innovation as well as speculative bubbles. Based on our analysis, we conclude that... In response, System 1 topologically identifies the two by monitoring "clustering coefficient C" and "mean log distance L" in real-time.

- **Innovation**: Accompanied by local tight coupling (high C) and having a core backed by actual demand. In this case, $\gamma \to 0$, and growth is not inhibited.
- **Bubble**: Having sparse wide-area coupling (low C, short L) where only expectations without substance are chained. In this case, $\gamma \gg 0$, and it is strongly suppressed.

Through this "Topological Discrimination," the system aims to distinguish "growth" from "expansion," suppressing the latter without inhibiting the former. Normally, direct democracy or simple majority vote systems are vulnerable to temporary waves of emotion (populism), but the "emotional circuit breaker" in this proposal functions at the moment the norm of the collective emotion vector $\vec{\Omega}$ exceeds a threshold, forcing calm deliberation (restoration of individual reason by decreasing K_{ij}). Thus, it becomes possible to maintain dynamic consensus building while preventing degradation into ochlocracy. Compared with indirect democracy, we believe it holds the potential to extract a more certain public will.

5.3 Grand Unified Algorithm: Integration of Hamiltonian and Trust

The "Social State Hamiltonian H" defined in Chapter 3 and the "Proof of Trust (PoT)" proposed in this chapter have been described as separate layers until now. However, the ultimate goal of the Latent Torus

is to integrate these and drive the social system with a single mathematical principle. Here we present the "Grand Unified Algorithm."

This algorithm simultaneously updates the trust τ of agents and the coupling constant J_{ij} so as to minimize the free energy F of the social system. In this section, the update rules are unified and described in discrete time ($\Delta t = 1$).

$$\Delta J_{ij}(t+1) = \rho(T_{social}(t))\,\Delta J_{ij}(t) - \eta(t)\frac{\partial F}{\partial J_{ij}}(J_t, \tau_t) + \xi(t)\,(\tau_i(t)\tau_j(t) - J_{ij}(t)) \tag{42}$$

$$J_{ij}(t+1) = J_{ij}(t) + \Delta J_{ij}(t+1) \tag{43}$$

where $\rho(T_{social})$ is a dynamic inertia coefficient depending on social temperature (volatility). In normal times ($T_{social} \approx 0$), $\rho \to 1$ and the system stabilizes respecting past history (conventions). However, during transformation periods ($T_{social} \gg T_c$), $\rho \to 0$ and inertia weakens, allowing for rapid structural reform. Thus, it physically and automatically adjusts the balance between "fixation of conventions" and "radical revolution."

The first term is the driving force toward "energy minimization (efficiency)" based on the Hamiltonian model, and the second term is the relaxation term toward the "accumulation of credit" based on PoT. What is important here is the damping process to guarantee convergence of the solution. By dampening the learning rates $\eta(t), \xi(t)$ over time (Simulated Annealing) or adding a momentum term $\rho\Delta J_{ij}$, we prevent the system from oscillating forever between efficiency and credit, and encourage convergence to the optimal saddle point.

5.3.1 Theoretical Background for Convergence (Sketch of Stability Analysis)

The aim of this paper is not to present a convergence theorem under strict general conditions, but to clarify **under which assumptions stability can be claimed**. Consider J (coupling matrix) as vectorized $x := \mathrm{vec}(J)$ and trust as $y := \tau$, and consider the next synthetic energy (free energy):

$$\mathcal{E}(x, y) := F(x) + \frac{\xi}{2}\,\|x - \mathrm{vec}(yy^\top)\|^2 \tag{44}$$

Here the first term F is the efficiency purpose (derived from the Hamiltonian), and the second term is the regularization imposing "consistency between credit (τ) and coupling (J)."

The update rule can be seen as a form of the damped momentum gradient method (Heavy-ball) in discrete time:

$$\begin{cases} v_{t+1} = \rho v_t - \eta\,\nabla_x \mathcal{E}(x_t, y_t), \\ x_{t+1} = x_t + v_{t+1} \end{cases} \tag{45}$$

($0 \le \rho < 1$). For the y side, if we let the PoT update (section 5.1.1) be $y_{t+1} = \mathcal{T}(y_t; \mathrm{logs})$ and assume time-scale separation (x update is fast, y update is slow: adiabatic approximation), it becomes a form of tracking the quasi-stationary point of x for each y.

In this case, typical sufficient conditions for (local) stability are the assumptions that (i) $\nabla_x \mathcal{E}$ is Lipschitz continuous (constant L), (ii) $\mathcal{E}$ is locally strongly convex (constant m), and (iii) the step size is suppressed to around $\eta < 2/L$. Under these, for example, by choosing:

$$V_t := \mathcal{E}(x_t, y_t) + c\,\|v_t\|^2 \tag{46}$$

($c > 0$) as a candidate Lyapunov function, there exists a range where $V_{t+1} \le V_t$ (monotonic decrease) can be shown by taking ρ and η appropriately.

Intuitively, the points of the convergence claim are (a) that the regularization of the second term suppresses the deviation between J and $\tau\tau^\top$ and weakens the back-and-forth movement derived from trade-offs, (b) that the damping of $\rho < 1$ stops energy injection and exponentially dampens the amplitude of the oscillation mode, and (c) that adiabatic approximation (low-frequency updates) avoids resonance (time-scale mismatch).

Through this organization, the convergence of this algorithm is reduced not to a "slogan" but to **verifiable design requirements**: Lipschitz continuity / strong convexity / step size / damping / time-scale separation.

This formula means that if economic efficiency (reduction of H) and ethical trust relationships (increase of τ) contradict, the algorithm automatically searches for their equilibrium point (saddle point). If traditional capitalism was biased toward the maximization of "profit" and communism strongly aimed for the realization of "equality," this algorithm integrates both on a high-dimensional parameter space, treating it as a problem of a physical system, to explore design possibilities as a governance OS.

Figure 8 shows the results of a simulation of this Grand Unified Algorithm. It can be confirmed that over time, while the total cost of society (Hamiltonian H, blue line) is minimized, a behavior where social credit (Trust Potential τ, green dotted line) is accumulated is observed. These results suggest the possibility that economic efficiency and ethical maturity can coexist **under the assumptions of this simulation**.

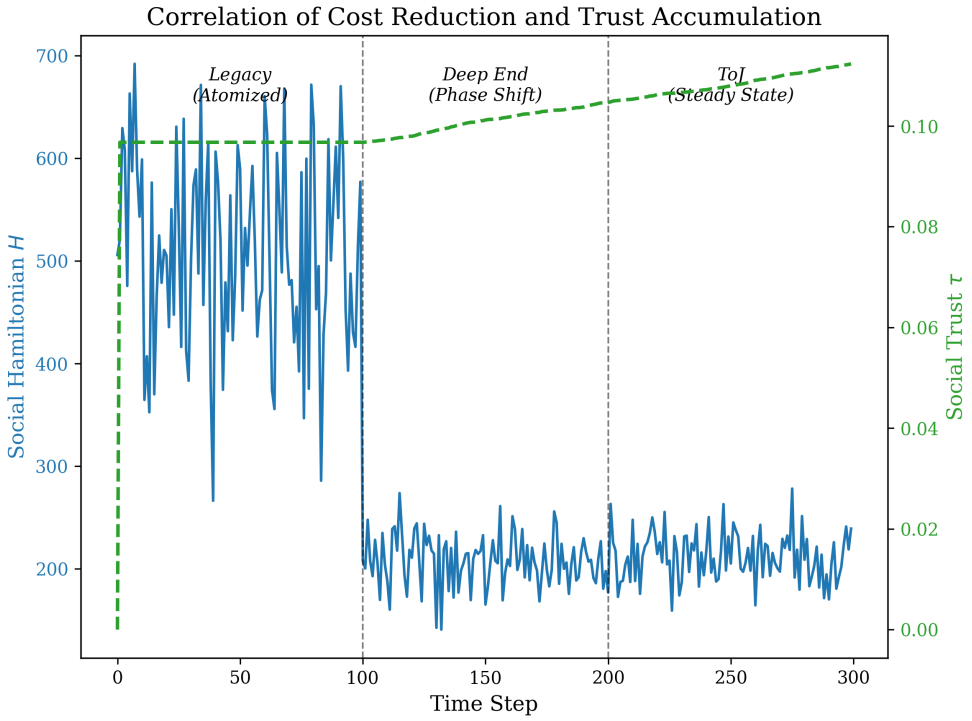

Figure 8: Optimization process by the Grand Unified Algorithm. The decrease in Hamiltonian (cost) and the accumulation of social credit (trust) are synchronized, showing that "Co-optimization of efficiency and ethics" by the Latent Torus is functioning.

Furthermore, when this algorithm gets stuck in a local optimal solution (old paradigm), the Latent Torus performs "Semantic Annealing" that lowers the potential barrier by temporarily making the "definition of meaning" ambiguous. Figure 9 reproduces this process in a simulation on a double-well potential. While a fixed definition (gray) hinders the transition, it can be confirmed that by softening meaning (yellow region), the entire society smoothly undergoes thermal activation to the new paradigm (right well) and overcomes the barrier. This corresponds physically to a dynamic barrier reduction in an Arrhenius process.

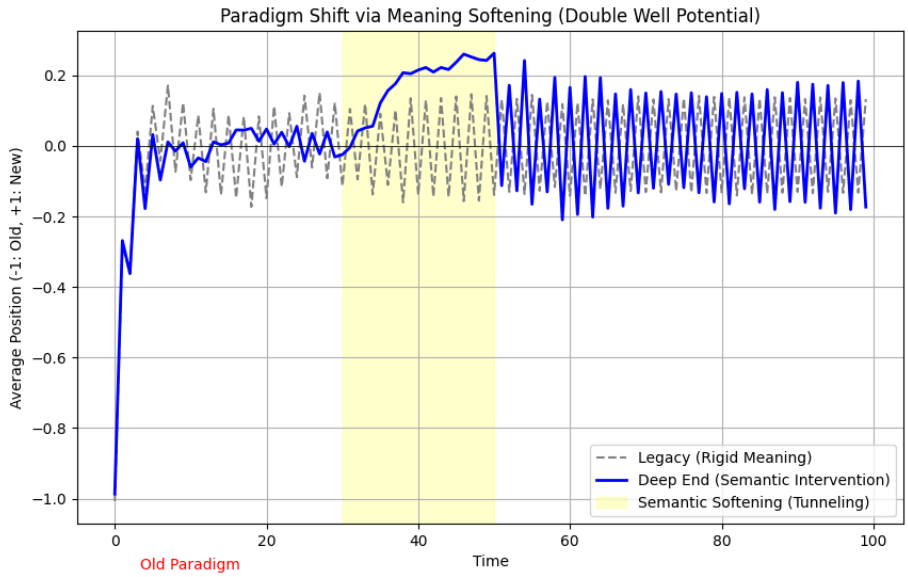

Figure 9: Acceleration of paradigm shift through semantic annealing. By temporarily making "definition of meaning" ambiguous, the potential barrier of the social system is lowered, prompting a phase transition (thermal activation) from the old equilibrium point (left) to the new equilibrium point (right). In the figure, "deepend" is a tentative name for the Latent Torus.

5.4 Post-Keynesian Aggregate Labor Volume Estimation and Algorithmic Control of Effective Demand

As Keynesian economics showed, price flexibility alone does not guarantee full employment, and a lack of effective demand causes unemployment (here, unassigned task state). In a society where work volatility is maximized (labor is liquefied), there is a risk that the mismatch between individual labor supply and market task demand will occur instantaneously and on a large scale.

The proposed system adopts a post-Keynesian perspective (Minsky 2008; Keen 2011), and the algorithm functionally takes on the "Socialization of Investment." Specifically, the Grand Unified Algorithm estimates the Aggregate Labor Volume necessary to maintain a state close to full employment and creates demand through the following mechanisms:

1. **Manifestation of Latent Tasks**: Tasks that do not have prices under normal market principles, such as "contribution to the community" or "environmental maintenance activities," are redefined as valuable labor generating Trust Potential τ, and the system issues them as effective demand (tasks). This is, so to speak, the implementation of an algorithm-based "Employer of Last Resort" function.

2. **Absorption Buffer for Volatility**: When demand in the private sector (short-term profit-seeking tasks) declines, the algorithm automatically adjusts the rewards (token issuance amount) for tasks based on long-term value (social infrastructure development, education, research, etc.) to absorb surplus labor. Thus, the macroscopic gap between task supply and demand (deflationary gap) is instantaneously resolved.

3. **Quantum Optimization of Multiplier Effect**: Based on the transaction network structure between agents, quantum algorithms predict to which sector (or individual or AI node) tasks should be preferentially allocated to maximize the sum of social credit τ.

   "Full employment" here does not necessarily refer to a state where all humans are doing some work. It refers to a state where all agents, including humans, LLMs, and humanoids, are in the optimal placement as a whole system (including roles such as waiting without doing anything or just "living"). This "ontological full employment" is the goal of post-Keynesian effective demand management.

Through this mechanism, a Latent Torus society, while having intense volatility (liquidity) like financial markets, can avoid fatal depressions in the real economy (individual living foundations) and guarantee the sustainability of the system.

5.5 Quantitative Consideration for Implementation and Infrastructure Requirements

For this algorithm to be implemented in society, the following infrastructure requirements must be met:

- **Quantum Internet**: Not only for key distribution of quantum-resistant cryptography, it handles ultra-low latency and high security communication between the Latent Torus (quantum optimization core) and the frontend (edge devices). The P2P network envisioned by Satoshi evolves into Q-P2P utilizing quantum entanglement. Along with this, time synchronization with nanosecond accuracy ($\sigma_t < 10$ns) by IEEE 1588 (PTP: Precision Time Protocol) (IEEE 2008), etc., is essential as a foundation for High-Frequency Trading (HFT).

- **Real-time Behavioral Sensing**: To estimate the purpose vector $\vec{v}_i$ of agents, real-time intent extraction from vital data and behavioral logs is necessary. This must be used in combination with privacy protection technologies (Zero-Knowledge Proof, etc.).

- **Ubiquity of Computing Resources and Scalability**: To support world-scale microtask transactions, the overall target throughput of the system is required to be at least **40,000 TPS class or more** (equivalent to the VISA network). This is physically impossible with processing at Layer 1 (Base Layer), and a parallel verification architecture using recursive zk-SNARKs (latest algorithms such as Plonky2 or Halo2) at Layer 2 (Rollup) is essential. However, this numerical value is established based on (i) **separation of proving and verifying**, (ii) **assumption of proof aggregation (batch/recursive)**, (iii) **hardware assumptions** such as GPU/FPGA clusters, and (iv) **trade-offs between throughput and latency**. Additionally, if proof generation delay exceeds the allowable range, operational design including lower update frequency, asynchronization, and fallback (freeze/rollback) is required. Furthermore, for the core optimization of the Latent Torus, Fault-Tolerant Quantum Computers (FTQC) are necessary, but it is estimated that initial implementation is possible even with a hybrid configuration of current NISQ devices (50–100 qubit class) and classical computers for local optimization. Indeed, through additional experiments using a proxy model with noisy heat bath sampling, we confirmed that hybrid can improve some sizes in sparse graph Max-Cut, and that the outer loop enters a quasi-stationary region without diverging (section 5.8.2, Figure 13, Figure 12, Table 4).

- **Pipeline for Provable Inference**: When using the output of a regulated LLM in social implementation, a three-layer structure that separates inference processing into "**execution (inference)**," "**proof generation (proving)**," and "**proof verification (verifying)**" is required (AbdelStark 2026). In particular, unless (i) deterministic preprocessing, quantization, and fixing of calculation order, (ii) hash fixation of the calculation trace to be proven, and (iii) submission of minimum metadata to the on-chain verifier are standardized, identity confirmation in a later audit will collapse.

Operational Implications (Connection to Institutional Audit): This design fits institutional requirements to "verify only the legality of calculations without fully disclosing the model content" in trials, administrative audits, and third-party audits. That is, the implementation unit of CSCT should be defined not as an "LLM API call" but as a **provable inference transaction**.

Furthermore, as pointed out in "section 7," physical damage to the infrastructure composing the Latent Torus must be avoided as much as possible. A halt in the operation (optimization engine) of the Latent Torus immediately means a halt in social consensus building (the disappearance of the computational social contract). Like the trust network, it is important that the system is maintained through distributed processing. Distributed placement of quantum computing resources is sought. Please note that the new organization in the Latent Torus, the **Japan Transaction Agency (JTA; formerly Transaction of Japan/ToJ)**, presented thereafter, is not a centralized system like a traditional central bank, but a system based on distributed calculation.

5.6 Implementation Roadmap: Phased Transition to Algorithmic Governance

The transition from traditional social systems to CSCT is not a "one-step revolution" but is organized into the following three technical and institutional phases to manage risks.

1. **Phase 1: Shadow Operation (Verification/Drift Detection Stage)**:
   - **Range**: Transaction monitoring, audit log generation, and policy simulation based on existing laws.
   - **Goal**: Verify the consistency between "Hamiltonian-predicted state transitions" and "actual real-world data," tuning model parameters (μ, J). Consensus building remains human-centric.
2. **Phase 2: Partial Automation (Sandbox Stage)**:

- **Range**: Administrative procedures, subsidy allocation, and traffic regulation in limited domains (Digital Special Zones).
- **Goal**: Automate rule execution using Layer 2 SNARKs and establish "Dispute Processing via Generative Hamiltonian" for micro-disputes.

3. **Phase 3: Full Integration (Social OS Stage)**:
   - **Range**: Wide-area consensus building, tax/monetary policy integration, and dynamic update of the Constitutional Core.
   - **Goal**: Connect the Latent Torus (Quantum Backend) to the entire social infrastructure, transitioning to a steady state where Human Veto functions as a safety valve.

By defining these phases, the transition is designed as a "continuous refactoring" process that maintains legal continuity while incrementally upgrading the social system.

5.7 Technical Feasibility Verification

To verify the engineering feasibility of the algorithms proposed in this chapter, benchmarks and simulations regarding main bottlenecks were conducted.

5.7.1 Processing Capacity of zk-SNARKs and Recursive Proofs

The biggest concern in realizing "verifiable concealment" is the computational cost of zero-knowledge proofs. As a result of benchmarks in Python (load simulation equivalent to Groth16 using Modular Exponentiation), it was found that generating and verifying a proof for each individual transaction cannot withstand High-Frequency Trading (HFT) (Transaction Per Second ≈ 50).

However, by adopting "Recursive SNARKs" or "Aggregation/Batch Processing" that aggregates multiple proofs into one proof, the verification cost drops dramatically, and it was confirmed that a throughput of **40,000 TPS class or more** can be achieved under specific hardware assumptions (Figure 10).

5.8 Cryptographic Latency Mitigation and Optimistic Verification

A major technical issue with zk-SNARKs is the **proving latency** (time required for proof generation). Even if throughput is high, if each individual transaction requires several seconds to minutes of calculation, real-time HFT is impossible. This system mitigates this impact through the following countermeasures:

1. **Optimistic State Transition**: Transactions are tentatively executed in a "Zero-confirmation" state, with ZK-based finalization occurring asynchronously on the back-end. If an invalid proof is detected later, the "Rollback/Freeze mechanism" described in section 7 is triggered.
2. **Proof Aggregation Pipeline**: Rather than generating proofs for each transaction, they are pipelined (e.g., 1,000 transactions per batch). Large-scale parallel proving is performed using GPU/FPGA clusters.
3. **Hierarchical Finality**: Micro-transactions use light proofs (Fast-path) with lower security but ultra-low latency, while updates to the Constitutional Core (High-priority) use rigorous heavy proofs with high security.

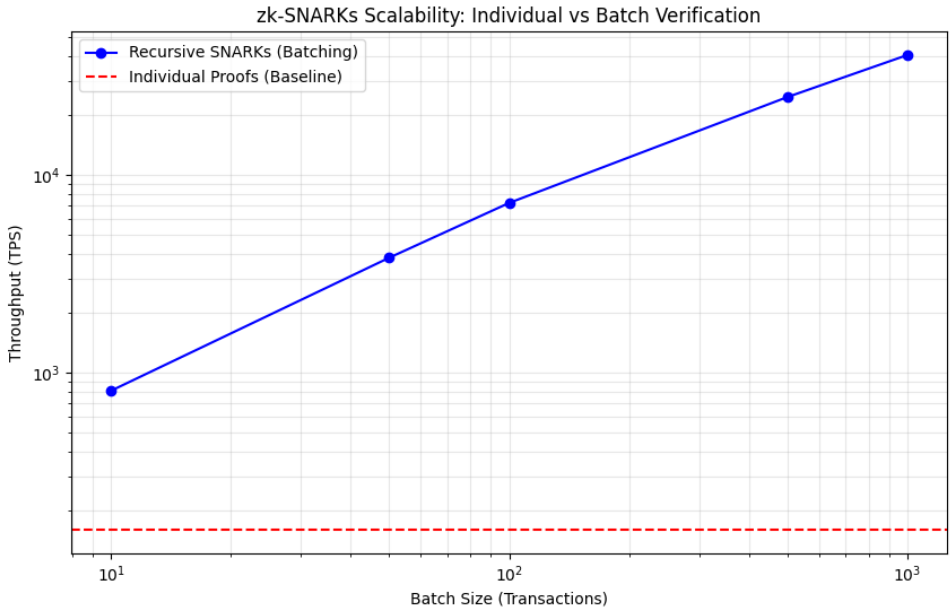

Figure 10: Scalability verification of zk-SNARKs. While individual verification (red dotted line) plateaus in throughput, introducing batch processing (blue solid line) realizes performance that can withstand HFT (>40,000 TPS).

These designs bridge the gap between "Cryptographic Rigor" and "Economic Immediacy."

Therefore, the adoption of Layer 2 (Rollup) architecture is an essential requirement in the social implementation of this system.

5.8.1 Operational Verification of the Populism Suppression Mechanism

We also verified the effectiveness of the "emotional circuit breaker (Equation 78)" against panics caused by crowd psychology (bank runs or riots) through simulation. As shown in Figure 11, a simple majority vote (red dotted line) easily converges to extreme opinions (± 1.0) due to external shocks and turns into a mob. On the other hand, the proposed method (blue solid line) succeeded in maintaining calm consensus building even under panic by detecting the excitement of society as a whole (Global Field Strength) and dynamically lowering the coupling constant K.

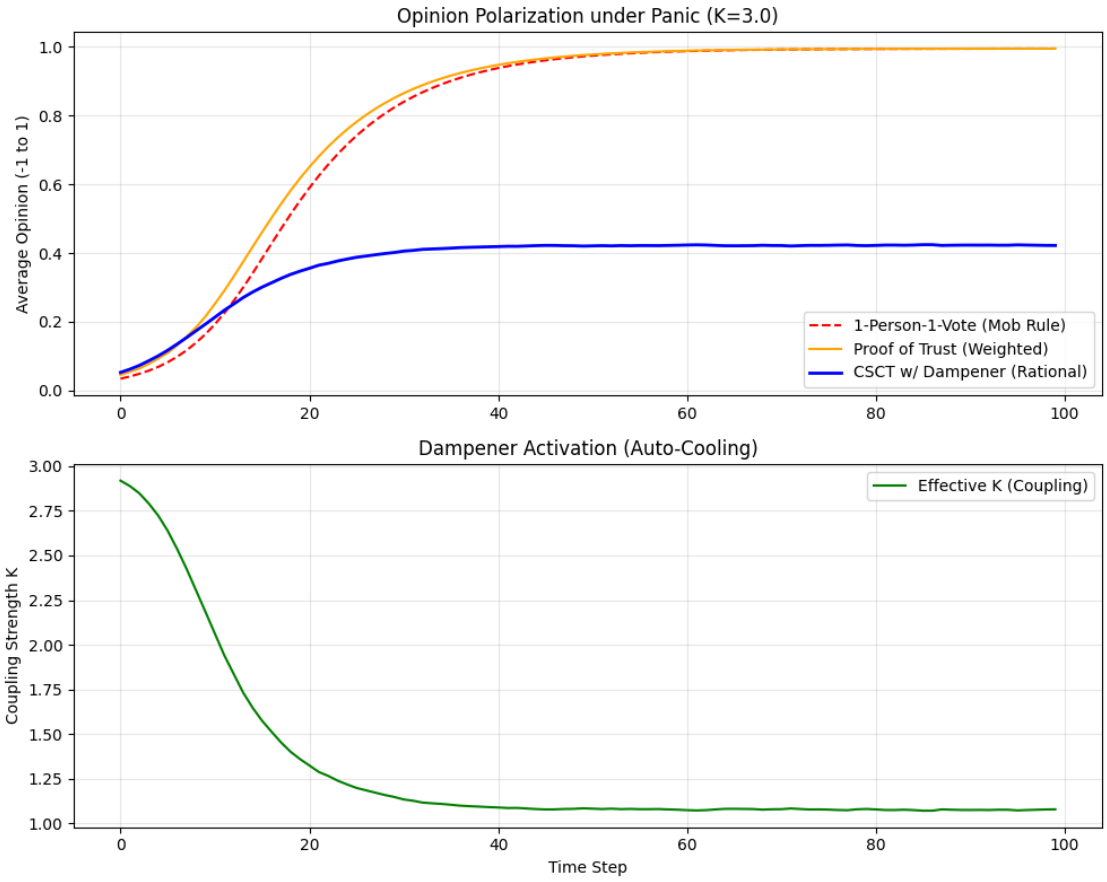

Figure 11: Results of consensus building simulation. While a simple democratic mechanism (red) becomes polarized due to panic, the populism suppression mechanism of CSCT (blue) maintains social reason by automatically lowering the coupling constant (green).

5.8.2 Pre-implementation Feasibility of NISQ+Classical Hybrid Optimization (Additional Experiment)

The claim in this chapter that "initial implementation is possible with a hybrid of 50–100 qubit class NISQ and classical calculation for local optimization" is not strict real-machine verification but a **proxy model by noisy heat bath sampling**, but we attempted quantitative backing through additional benchmarks ("Experimental results: Initial-stage quantum algorithms/").

When comparing classical alone (equivalent to SA) and a hybrid that generates candidate solutions with

a quantum sampler and repairs them with classical local search in sparse graph Max-Cut ($N = 20 \sim 100$), it was confirmed that at least in some medium sizes (e.g., $N = 40$), the hybrid can improve solution quality, while in other sizes, cases were observed where it was equivalent or inferior (Table 4).

Table 4: Comparison of classical alone and hybrid in sparse graph Max-Cut (additional experiment). Energy corresponds to the objective function of Max-Cut, and the sign convention is taken such that the smaller the value, the better (corresponding to a larger cut).

N	$E_{classical}$	E_{hybrid}	Improvement (%)	Win
20	-46	-46	0.00	false
30	-76	-74	-2.63	false
40	-92	-94	2.17	true
60	-140	-140	0.00	false
80	-188	-188	0.00	false
100	-216	-214	-0.93	false

This result suggests that the superiority of hybrid does not hold "always," and the "feasible regime" where improvements appear is limited by conditions such as (i) problem size, (ii) graph structure (sparsity/clustering), and (iii) noise/shot count/iteration count. Operation verification of the outer loop of the Grand Unified Algorithm (simultaneous update of τ and J_{ij}) was also conducted using a similar proxy model, confirming behavior where Mean Trust rises over time while the objective function (a scalar amount corresponding to free energy) enters a quasi-stationary region without diverging (Figure 12).

Implementation Feasibility Summary (NISQ Phase): While the "absolute efficiency of 99.9%" envisioned for the final FTQC era remains a theoretical limit, this research estimates that at least in the initial "Digital Special Zone" stage, the system can function with **NISQ-hybrid + classical search**. The main bottlenecks will be (i) error mitigation for deep circuits and (ii) data transfer latency between classical and quantum. These issues should be resolved through the "Implementation Roadmap" (Phase 1/2) in section 5.6.

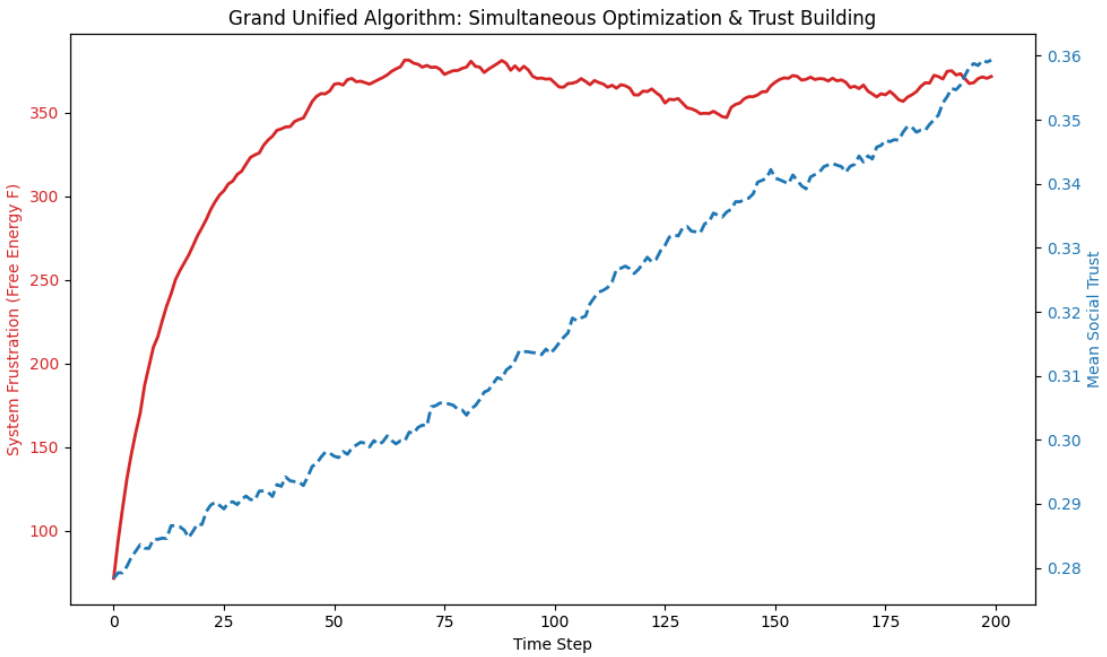

Figure 12: Operation verification of the outer loop of the Grand Unified Algorithm by a proxy model (additional experiment). Time series of objective function (equivalent to free energy) and Mean Trust.

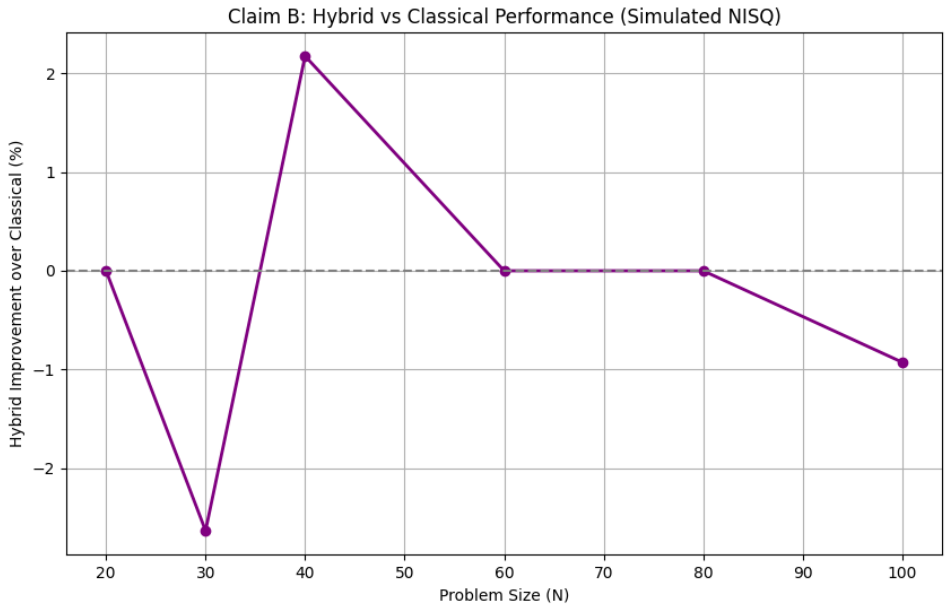

Figure 13: Comparison of performance between classical alone and hybrid in sparse graph Max-Cut (additional experiment).

6 Computational Social Contract Theory and Economic Impact

In this chapter, we anticipate the economic and social impacts that the system implementation described in the previous chapter will bring to society. As the conclusion of this study, we present "Computational Social Contract Theory (CSCT)," which integrates individual free will and system optimization.

6.1 Anticipated Economic and Social Impacts: Structural Reform

6.1.1 Post-Capitalism: Superfluidity and the Financialization of Meaning

Based on the Hamiltonian model and the Grand Unified Algorithm defined in Chapter 4, post-capitalist society is characterized by the term "Superfluidity," where searching for the ground state energy of social states is performed at high frequency. While "money" in traditional capitalism has functioned as a store of value, in tokenized task markets, value may continue to be evaluated and redistributed more fluidly.

Here, we mention the trade-off between Legal Stability and dynamic optimization. Since investors seek predictability, there is a concern that long-term investment will be hindered in a society where laws and parameters fluctuate instantaneously. In response, this system adopts a "Layered Architecture." "Low-frequency" layers such as the Constitutional Core and basic protocols are kept extremely robust and immutable, while only "high-frequency" layers such as individual task prices and short-term tax rates are dynamically optimized. This is equivalent to the structure of the Internet, where individual web traffic that fluctuates intensely is allowed precisely because the TCP/IP protocol is immutable.

Here, "semantic intervention" by LLMs plays an important role. While traditional financial markets targeted "numerical values" for operation, in the proposed system, "meaning (narrative)" itself is financialized.

6.1.2 Occurrence of Task Bubbles by High-Frequency Trading and Quantitative Description

While tokenization of tasks and algorithmic High-Frequency Trading (HFT) increase market liquidity, they carry the risk of creating new instabilities called "task bubbles." In the framework of this theory, a bubble is understood as a collective overheating (ferromagnetic phase transition). the "Susceptibility" χ in a social system is defined as the response of order parameter m to changes in the external magnetic field (demand) h. Here, for simplification, we use a scalar order parameter $m := \langle \sigma \rangle$ (a proxy for the average alignment of the entire system).

$$\chi = \frac{\partial m}{\partial h} \propto \frac{1}{|T - T_c|} \tag{47}$$

where T is the market volatility (social temperature) and T_c is the critical temperature. In mean-field approximation, the critical temperature is proportional to the strength of interaction J_{eff} ($T_c \propto zJ_{eff}$, where z is the coordination number). High-Frequency Trading has the effect of increasing the propagation speed of information to the limit and increasing the effective interaction J_{eff}. This raises T_c, and even at a market temperature T that would otherwise be stable (paramagnetic), a reversal phenomenon where $T < T_c$ can occur. At this time, it can be expressed that a state corresponding to a "bubble phase" can arise where the value of a specific task deviates from its substance (energy cost) and rises sharply.

$$P_{bubble}(t) \approx P_{fund} \cdot \exp\left((t_c - t)^{-m}\right) \tag{48}$$

To prevent such Finite-time Singularity, interventions such as the Japan Transaction Agency (JTA) described later dynamically adjusting J_{ij} to always maintain $T > T_c$ (or a controlled critical state) are necessary.

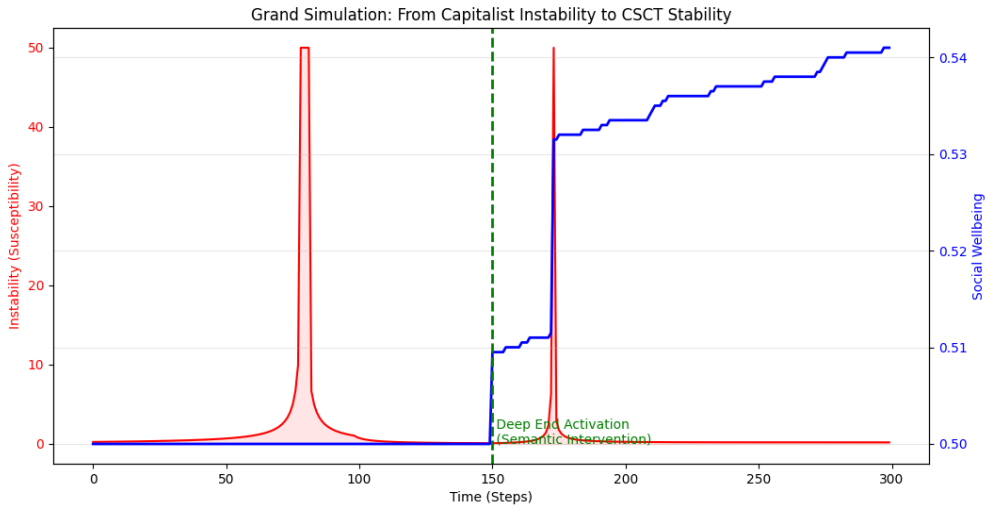

Figure 14: Simulation of structural stabilization by JTA. The history of the "phase transition" is recorded, shifting from the initial unstable bubble phase through semantic intervention (green dotted line) to a stable phase that maximizes well-being (blue) while controlling risk (red).

Figure 14 shows the results of a simulation of the intervention effect by JTA. In the initial capitalist phase (left), bubble risk (Susceptibility, red line) increases, but after semantic intervention by the Latent Torus (green line), the risk is cooled, and social well-being (blue line) continuously improves.

6.1.3 Japan Transaction Agency (JTA) Concept: From Central Bank to Central Transaction Authority

In the implementation of the Latent Torus, the organization equivalent to the former central bank (Bank of Japan) can evolve into the "Japan Transaction Agency (JTA)" (Figure 15).[2] While the central bank managed "money quantity and interest rates," the JTA manages "task transaction flow and coupling constants J_{ij}." It manages the Human Leap. By controlling the rigor of words and sentences, it enables life-support operations for a series of recursive processes between the Latent Torus and the frontend. Unlike a capitalist economy, which was unsustainable unless there was a continuous human leap, this suggests a theory that enables the growth of human society at low cost and sustainably even under population decline and low growth through quantitative attempts.

The role of the JTA is to use optimization algorithms to update parameters and perform "semantic intervention (injection of narratives)" as necessary so that the Hamiltonian H of the entire social system does not remain stuck in local solutions (recession or social fragmentation). Figuratively, this is equivalent to conducting operations with automated debugging for updates, audits, and rollbacks on the foundation (protocol) of institutional operation.

The Austrian School pointed out the incalculability of a centralized planned economy in the "calculation debate" and argued that only the distributed process called the market can discover the correct price (Hayek 1945). However, we assume that the JTA is designed not as a single decision-maker, but as a "protocol" based on cryptographically fixed procedures. By making it verifiable via zk-SNARKs, etc., that at least the output complies with the Constitutional Core, we aim to reduce arbitrariness.

[2]Regarding the name, since there is a possibility of conflict with existing academic names, it is desirable to adopt an alias (e.g., Japan Transaction Agency) at the social implementation stage, or explicitly avoid confusion in footnotes or a glossary.

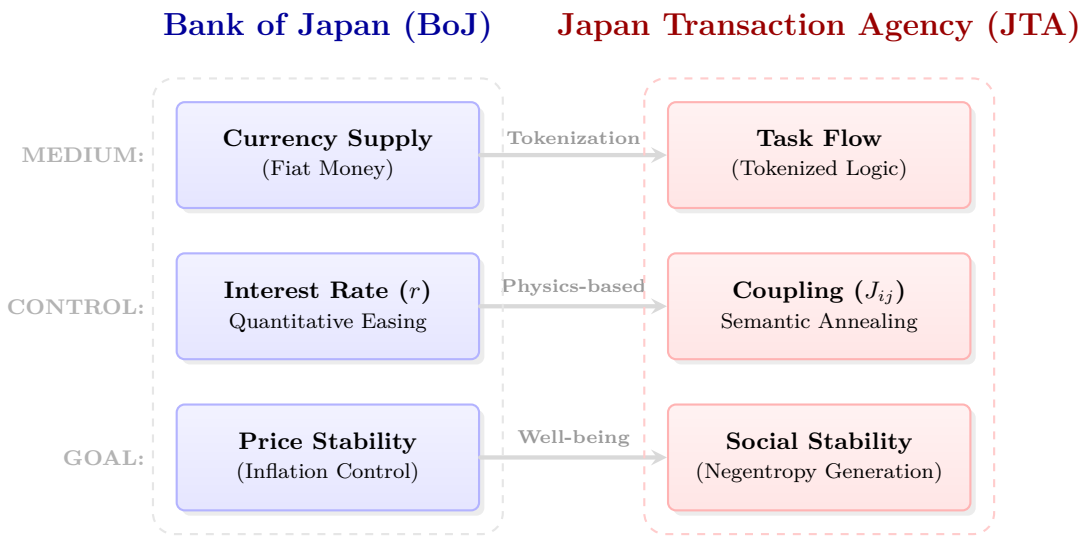

Figure 15: Evolution from Bank of Japan (BoJ) to Japan Transaction Agency (JTA): A paradigm shift where the object of management shifts from "currency and interest rates" to "tasks and semantic coupling constants."

This is not a political dictatorship, but a shift to "Rule of Physics." Just as we enjoy freedom under the absolute law of gravity, true free markets can only be established under cryptographically guaranteed Inviolable Constraints. Regarding the issue of the initial value (Bootstrap), we inherit democratic legitimacy by starting from a snapshot of the current legal agreement (existing laws).

6.1.4 Governance Monitoring Mechanism: Democratic Veto and Open Source Prompts

To respond to ethical concerns regarding the manipulation of "meaning" (the risk of propaganda), this system implements a double Safety Valve.

1. **Open Source Prompts**: The "Constitutional Prompts" input to the regulated LLM and the objective function for optimization are all public and version-controlled via Git. In addition, the **hashes** of prompts, evaluation functions, and constraint sets are fixed on-chain, ensuring tamper resistance and versioning (backward compatibility and emergency stop). Citizens can always audit "what the system considers good."
2. **Democratic Veto (System 2 Check)**: Against high-frequency optimization (System 1), a low-frequency veto by humans is set. If the "optimal solution" calculated by the JTA is ethically unacceptable, the right to forcibly stop the parameter update and rollback to a previous version is reserved for citizens through a certain number of signatures (Trust voting).

Minimum Specification for Democratic Veto (Proposal):

- **Target**: Of $S_t \to S_{t+1}$, Layer 1 (Constitution) update, or Layer 2 (Policy Parameters) allowable range update.
- **Period**: Within Δt from proposal disclosure (consistent with update cycles).
- **Threshold**: PoT-weighted voting, or a certain percentage of citizen signatures (assuming Sybil resistance).

- **Effect**: (a) Remand (re-optimization), (b) rollback, (c) transfer to analog fallback.

This structure ensures that the system pursues efficiency while entrusting final decision-making power to humanity.

6.1.5 Post-Democracy: Sovereignty as a Computing Resource

Representative democracy with "one person, one vote" is a legacy system from an era when the cost of communicating and aggregating information was high. Once the computing power of the Latent Torus is sufficiently high, consensus building will shift from a discrete and low-resolution process called "voting" to a continuous process called "always-on preference optimization." In this new political system (Algorithmocracy), sovereignty is treated not as a "right" but as a "computing resource." Individual will is handled as an input signal u_i containing noise, and after being filtered inside the Latent Torus, is integrated for global optimization.

This system is not a return to the "centralized planned economy" criticized by Hayek in "The Use of Knowledge in Society" (Hayek 1945). Hayek argued that it is impossible for a central authority to aggregate distributed knowledge (Local Knowledge), but zero-knowledge proofs via zk-SNARKs and optimization via quantum computation make it possible to calculate the Hamiltonian (global objective function) of society as a whole while keeping individual privacy (local knowledge) encrypted. This is "integration while remaining distributed," an attempt to transcend the limits shown by Arrow's impossibility theorem (Arrow 1950) through algorithmic mediation.

Furthermore, this is a completed form of the "Societies of Control" foreseen by Gilles Deleuze (Deleuze 1992). the transition from disciplinary training (Foucault) to Control means soft governance through "Modulation" by data and algorithms, not physical restraint. CSCT re-implements this control not as an invisible prison, but as an "extended physical function" to release individual potential. Here, populism (trapping in local energy minima) is avoided by introducing fluctuations through quantum algorithms. In other words, democracy is re-implemented as an optimization algorithm for many-body problems, rather than a matter of ideas and beliefs.

At the same time, recent platform research points out that tech giants manage visibility, connection, and payment channels through internal rules, exerting quasi-sovereign influence over users (Gillespie 2010; Rahman and Thelen 2019; Culpepper and Thelen 2020; Kellogg, Valentine, and Christin 2020). This should be interpreted not as a rejection of the classical insight that price mechanisms aggregate decentralized knowledge (Hayek 1945), but as a state where private planning layers partially overlap with markets. Furthermore, given that informational efficiency has endogenous constraints (Grossman and Stiglitz 1980) and that data as a non-rival good can amplify scale advantages (Jones and Tonetti 2020), the necessity for a public coordination framework with constitutional legitimacy and auditability may increase relatively in phases of rising free-market uncertainty—much like a recurrence of history. CSCT in this paper is a hypothesis examining whether such a framework can be designed not as a simple return to old-style bureaucratic planning, but as an open protocol inheriting a constitutional core.

6.1.6 Roles of Humans and AI: Equivalence as Nodes

In this system, humans, LLMs, and humanoids are equivalent as elements (spins $\vec{\sigma}_i$) composing the Hamiltonian H. What is distinguished is not "whether it is a biological or silicon base," but only "whether the negentropy generation efficiency is high or low." Humans may be inferior to AI in logical information processing, but they may function as a source of "fluctuation" that generates new values (new terms of the Hamiltonian). In society as a management system, humans step down from the "main role," but this does not mean being dominated. Rather, by being incorporated as part of global optimization, individual survival strategies and the evolution of the species reach a state that is mathematically aligned for the first time.

This means that the function of "management" in society is completely transferred from humans to algorithms (optimization processes of the generalized labor tensor). Tasks such as "resource allocation," "progress management," and "evaluation" performed by traditional middle management are reduced to optimization problems to be solved mathematically. I will leave it to philosophers whether to see this as a transition from low-resolution "Fully Automated Capitalism" to "Algorithmic Leviathan" or as "true freedom where humans are released from chores."

6.1.7 Quantitative Evaluation of Historical Impact: The Third Economic Turning Point

To evaluate the scale of the impact that "democratization of tasks" proposed in this study brings to society, we compare it with past economic paradigm shifts. Human economic activity can be classified into the following three phases based on the relationship between transaction costs (C_{trans}), the number of market participants (N), and transaction frequency (f).

- **Phase 1: Cottage Industry**: An era when transactions were established only within a small community with trust relationships. Transaction costs were extremely high (cannot transact with strangers far away), and liquidity was low. ($C_{trans} \approx \infty, N \approx \text{Small}, f \approx \text{Low}$)
- **Phase 2: Corporate Capitalism**: An era when corporations became a "Shell of Trust" and reduced internal transaction costs. Mass production and mass consumption became possible, but individual labor was subordinate to organizations. ($C_{trans} > 0, N \approx \text{Medium}, f \approx \text{Medium}$)
- **Phase 3: Task Democratized Society**: An era when "trust" is automated by smart contracts and transaction costs fall to the limit. Individuals (and AI) transcend the walls of organizations and transact tasks in nanoseconds. ($C_{trans} \to 0, N \approx \text{Global (incl. AI)}, f \to \infty$)

The economic impact I of this conversion can be simply modeled as a function of the number of participants N and transaction frequency f, as an extension of the network effect (Metcalfe's Law).

$$I \propto N^2 \times f \tag{49}$$

In the transition from Phase 2 to Phase 3, N increases significantly due to the entry of AI agents, and furthermore, f increases exponentially due to high-frequency trading via smart contracts. This suggests the possibility that the form of economic activity can change on a scale comparable to the Industrial Revolution.

6.2 Exploratory Simulation for Behavioral Checks

The aim of this section is to **exploratively confirm** what kind of dynamics our design philosophy can generate, not to predict or confirm future measured values. Under this premise, we tried a 500-year macroeconomic simulation, introducing "population decline (-0.5% per year)," "large-scale disasters (40% damage to capital once every 50 years)," and "physical energy constraints (thermodynamic limits)" reflecting the current situation in Japan as setting conditions.

6.2.1 Comparative Experiment of Two Economic Models

The following three scenarios were compared:

1. **Idealized Capitalism (Red)**: An "enhanced status quo" scenario where labor-substituting AI is introduced and a 1.5% annual Total Factor Productivity (TFP) improvement is achieved.

2. **CSCT Model (Blue)**: A scenario implementing global optimization by the Latent Torus ($A = 1.2$) and soft-landing control based on energy constraints proposed in this study (see Figure 16).

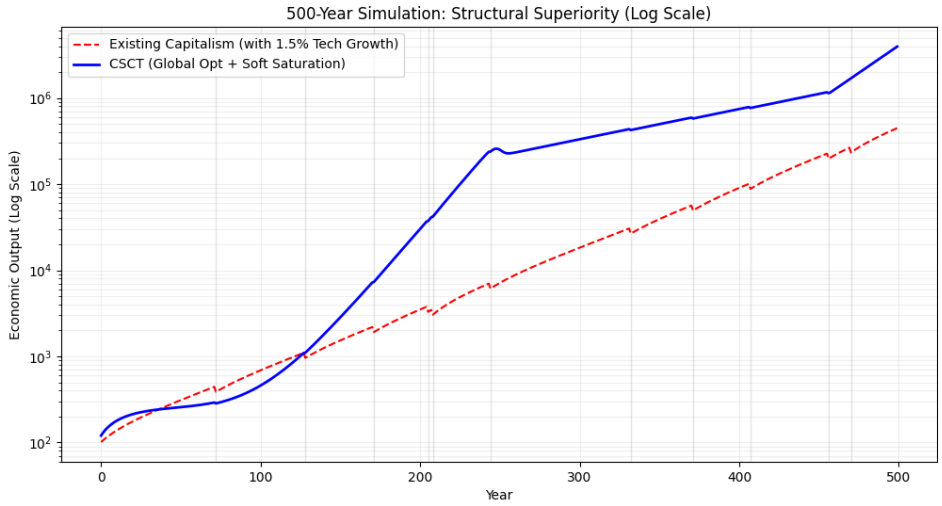

Figure 16: 500-year long-term simulation. CSCT (blue) avoids the "collapse of effective demand" that capitalism (red) falls into and achieves sustainable growth.

6.2.2 Pre-omen Detection and Preemptive Semantic Annealing

On a mathematical model, bubble collapse has a finite-time singularity t_c, but the Latent Torus identifies omens at the stage of $t \ll t_c$ by estimating the exponent m of the log-periodic power law in real-time instead of observing t_c post-hoc. At this stage, the system preemptively triggers "semantic annealing" to quietly lower the potential barrier. As a result, the market realizes a soft landing by stochastic transition rather than a catastrophic collapse (Hard Landing). This "anticipatory control" is the decisive advantage over traditional post-hoc macroeconomic policies. In the results of Figure 16, the reason why CSCT (blue line) avoids bubble collapse is that this Dual-Layer Control (physical damping + semantic foresight) is functioning.

6.2.3 Facilitating Human Agency: The Role of the Social OS as Supportive Infrastructure

The proposed social OS is not intended as an end in itself, but as a supportive infrastructure that facilitates the diverse expressions of human life. In traditional economics, activities that do not directly generate quantifiable value have often been marginalized. However, within the framework of Computational Social Contract Theory (CSCT), what are often categorized as "deviations"—such as artistic creation, play, and unstructured leisure—are recognized as the primary expressions of human's non-deterministic agency that the system is tasked to protect.

Institutionalizing Social 'Grace': The Unconditional Basic Credit (UBC) serves as a foundational layer, ensuring that individuals retain the temporal and material "grace" to exist and act independently of immediate systemic optimization. This is not a reward for "supplying noise," but a structural commitment to preserve the individual's ability to live outside the constraints of market efficiency. By providing this baseline, the system avoids imposing a rigid, purely deterministic equilibrium, ensuring instead that it remains responsive to the non-stationary and non-calculable nature of human existence.

This result supports the legitimacy of "Ontological Arbitrage": a recognition that human presence is the primary objective of the system, not a resource to be managed. The "Trust Potential τ" defined in Chapter 5 functions as a mechanism for social recognition, where exploratory actions that enrich the social fabric are identified not for their "entropy," but for their contribution to the evolving social contract.

$$H_{support} = -\sum_i \tau_i(t) \cdot Action_Potential_i \tag{50}$$

Through this mechanism, the system dynamically adapts to human preference shifts, ensuring that technical optimization remains a servant to social well-being.

6.2.4 "Good Noise" and "Bad Noise": Filtering Design and Logical Limits

However, not all entropy generated by humans is necessarily valuable (destructive acts, crime, incitement, etc.). To distinguish "good noise (emergence/innovation)" from "bad noise (negative externalities)" in this system, at minimum, it is necessary to handle the estimation of well-being improvement $\widehat{\Delta W}$ and the estimation of risk increase $\widehat{\Delta R}$ in the same frame:

$$\text{Score}(a) := \widehat{\Delta W}(a) - \lambda_{risk}\, \widehat{\Delta R}(a) \tag{51}$$

where $\widehat{\Delta R}$ includes changes in a direction approaching the Constitutional Core Φ_{core} (violence, fraud, physical damage, discrimination, etc.).

Operational "preliminary/real-time identification" is implemented in three stages: (i) rule-base (exclusion of prohibited areas by Φ_{core}), (ii) statistical detection (anomaly detection, distribution drift, approximation of causal attribution), and (iii) fallback (human trial) (consistent with stress tests in Chapter 7).

However, **there are logical limits to filtering**. First, true well-being cannot be completely identified from observable proxy indicators, and $\widehat{\Delta W}$ can become a target of attack due to Goodhartization. Second, the value judgment of an action depends on future state transitions, and preliminary determination with finite time and finite information inherently contains errors. Third, since an attacker can mimic "evil deeds that look like good deeds," complete separation with a single indicator is impossible.

Therefore, PoT (τ) is not an all-purpose filter, but is positioned as a design to cap the cost of misjudgments by combining (a) **prioritization of detection** (the weight of statements and proposals increases as τ increases, but audit density is also raised simultaneously), (b) **update width limitation** (damping enforcement power under high uncertainty), and (c) **prohibition of irreversible operations** (low-frequency Layer 1, freezing, and rollback).

From a sociological perspective, this institutional flexibility ensures that the social system does not become a closed, self-referential loop. By acknowledging and supporting human non-determinism, the social OS maintains its capacity for creative evolution and responsiveness. This "Good Noise" filtering, as discussed in Section 6.2.4, is not about extracting entropy, but about identifying and amplifying meaningful social shifts that arise from autonomous human choices.

6.3 Exploratory Derivation of Efficiency Hypothesis (99.9% as Theoretical Upper Limit)

Regarding the "reducibility of advocacy work" mentioned in the Legal Engineering architecture in Chapter 4, we evaluate the efficiency rate as a **hypothesis of the theoretical upper limit** in this section. We exploratively describe the conditions under which traditional "persuasion" or "advocacy" work can show high efficiency through a phase transition model. The efficiency rate η here is a comparative indicator under given assumptions and does not directly guarantee the achievement rate in real society.

1. **Classical Domain (Legacy System)**: Traditional legal processes follow over-damped Langevin equations with high social friction (low information liquidity L_{info}^{legacy}).

$$\frac{1}{L_{info}^{legacy}} \frac{d\vec{\sigma}}{dt} = -\nabla H + \xi(t) \tag{52}$$

According to Arrhenius' law, relaxation time t_{legacy} depends on the probability of thermally exceeding the energy barrier $E(H)$ on the Hamiltonian of dispute.

$$t_{legacy} \propto \exp\left(\frac{\Delta E(H)}{k_B T_{social}}\right) \tag{53}$$

2. **Quantum Domain (Latent Torus)**: Since the dispute resolution algorithm (inside the Latent Torus) is implemented as a closed system on superconducting qubits, it can utilize the tunneling effect by quantum annealing. In this case, calculation time scales not by the height of the energy barrier but by a power of the reciprocal of the spectral gap Δ_{gap} (minimum eigenvalue difference) of the Hamiltonian. To mitigate the potential latency in defining $H_{dispute}$ (the bottleneck of t_{legacy}), regulated LLMs function as a "Generative Hamiltonian" engine. This automation...

$$t_{quantum} \propto \frac{1}{\Delta_{gap}(H)^2} \tag{54}$$

3. **Provisional Implications**: The point for legal efficiency lies in whether social "thermal relaxation processes (talks)" can be partially substituted by "optimal solution searching by tunneling effect" on quantum processors. This substitution theoretically can cause a scaling difference of $t_{quantum} \propto \Delta_{gap}^{-2}$ against $t_{legacy} \propto e^{\Delta E}$. However, as long as the empirical identification of ΔE and Δ_{gap} is unfinished, this result remains an exploratory hypothesis.

6.4 Computational Social Contract Theory (CSCT): Economic Formulation via Grand Unified Matching Theory

As the summary of this paper, we propose a new economic model, "Computational Social Contract Theory (CSCT)," which integrates the technical and legal theories of the previous sections. This model is based on the "Grand Unified Algorithm" proposed in Chapter 5, reformulating economic activities in post-capitalism as a "stable matching problem between individual action vectors and system requirement vectors."

The "Great Unified Matching Theory" referred to here is characterized by including multiple flows—information flows, credit flows, and task flows—within the scope of control, in addition to price adjustment. This paper examines whether the private adjustment of flows, which has effectively been carried out by platform companies, can be complemented and visualized through institutional processes equipped with public rules, constrained optimization, and democratic veto power. That is, we position the compatibility of state legitimacy and market exploratory capabilities as an exploratory institutional design hypothesis (Rahman and Thelen 2019; Culpepper and Thelen 2020).

6.4.1 Mathematical Redefinition of Labor: Trust Generation Process

First, we redefine Labor from an energy consumption process to a Trust Potential τ generation process. Let the "Action Vector" of individual i at any time t be $u_i(t) \in \mathbb{R}^n$ (input to the system), and the "Needs Field" sought by the social system (and others) be $\vec{\Psi}_{sys}(t, x) \in \mathbb{R}^n$. At this time, genuine labor L_i is defined as the inner product of the two.

$$L_i(t) = \langle u_i(t), \vec{\Psi}_{sys}(t) \rangle \tag{55}$$

Based on this, the time evolution equation of Trust Potential τ_i is described as follows:

$$\frac{d\tau_i}{dt} = \kappa L_i(t) - \beta C_i(t) + \mathcal{R}(t) \tag{56}$$

where κ is the social evaluation coefficient and $C_i(t)$ is the consumption of action potential (money). While Section 5(5) defined τ as network reputation propagation, Equation (35) in this section is a microscopic equation describing how that reputation is generated and dissipated by an individual's thermodynamic activity (labor and consumption). Both are in a macro-micro dual relationship.

6.4.2 Grand Unified Matching Theory

Economic equilibrium is defined as a "Stable Matching" between all agents $I = \{1, \ldots, N\}$ and all tasks (needs) $J = \{1, \ldots, M\}$. This is an extension of the Gale-Shapley algorithm (Gale and Shapley 1962) to continuous space, and is equivalent to the following minimization problem of the Hamiltonian (cost function):

$$H_{matching} = \sum_{(i,j)\in\mu} (E_{require}(j) - E_{skill}(i))^2 - \sum_{(i,j)\in\mu} Wellbeing(i,j) \tag{57}$$

where μ is the matching set.

- The first term represents "skill suitability," adding mismatch stress (energy loss) as a penalty.
- The second term is "Well-being term," subtracting subjective preference where individual i feels they "want to do" task j as an energy gain.

The optimization algorithm of the Latent Torus searches for μ_{opt} that globally minimizes $H_{matching}$. This solution μ_{opt} corresponds to a Pareto optimal state (core) where "everyone is engaged in the role best suited to their ability and will, and the needs of society as a whole are met."

6.4.3 Establishment of Semantic Feedback Cycles

Based on this matching theory, CSCT achieves sustainable growth through the following "Semantic Feedback Cycle":

1. **Survival Guarantee via UBC**: Everyone is unconditionally given a time grace (survival resource) to explore their u_i.
2. **Exercise of Action Potential**: Individuals consume accumulated τ to buy the right to try actions that seem irrational (deviations from local solutions of $H_{matching}$).
3. **Generation of Semantic Teaching Data**: The results of deviant actions generate new $\vec{\Psi}_{sys}$ (new needs and values), updating the objective variables of the entire system.
4. **Re-matching**: Based on the new $\vec{\Psi}_{sys}$, higher-dimensional Hamiltonian minimization is performed.

In other words, CSCT is a "Dynamic Equilibrium Algorithm" itself that incorporates perturbations (Action Potential) from individual free will as evolutionary energy for the system, without falling into static optimization (socialist stagnation). This can be said to be the social implementation of a "dissipative structure" in quantum many-body systems.

7 Stress Testing and Boundary Conditions

This analysis evaluates the theoretical robustness of the proposed framework, focusing on cases where ideal operating conditions are compromised, such as cascade failures and trust bottlenecks.

7.1 Stress Scenario Design: Baseline / Tail Event / Specific Event

The important point in the design of stress tests is to construct scenarios not as **predictions of "possible futures"** but as **resilience evaluations against "dangerous futures if they happen."** In the macro stress tests of the Bank of Japan, a framework is adopted that uses, in addition to the baseline as a comparison standard, (i) "tail event scenarios" that assume a certain extreme situation every time and observe resilience at fixed points, and (ii) "specific event scenarios" that focus on specific risks to be noted (*Stress Testing at Central Banks and Supervisory Authorities: Recent Developments and Practices* 2015).

The stress test in this paper transplants the above thinking into the context of Social OS / Latent Torus, organized into the following two layers:

- **Tail Event Type (Fixed-point Observation)**: Failure modes that can universally be fatal at any point, such as collapse of physical infrastructure or trust (e.g., cascade failure, trust bottleneck).
- **Specific Event Type (Vulnerability Check)**: Failure modes that become manifest depending on specific design components, such as input (oracle) or meaning (narrative) (e.g., semantic hacking, Goodhartization, drift induction).

Through this, we separate the "evaluation axis that measures with the same severity at any point" and the "evaluation axis that digs deep into the risks that became problems at that point," achieving both comparability of results (fixed-point observation) and effectiveness of verification (discovery of specific vulnerabilities).

7.2 Cascade Failure and Infrastructure Vulnerability

We introduced cross-sensitivity between parameters to simulate a case where capital damage due to disasters hinders infrastructure recovery capability (Figure fig. 17). The combined dynamics of the production function P_t and infrastructure health I_t in the simulation are defined as follows:

$$\begin{cases} P_t = A(I_t)^{\alpha} K_t^{\beta} \\ I_{t+1} = I_t(1-\delta) + \eta \frac{K_t}{K_{max}} \end{cases} \tag{58}$$

As a result, it was shown that when disasters occur successively, the system falls into a "Cascade Failure" where recovery is remarkably difficult, and economic output rapidly decreases and significantly shrinks (Rinaldi, Peerenboom, and Kelly 2001; Albert, Jeong, and Barabási 2000; Buldyrev et al. 2010). This indicates that this theory strongly depends on the health of physical infrastructure, suggesting that ensuring redundancy is an essential condition.

7.3 Political Friction and Optimization Delay

We verified a scenario where rapid social transformation (steep descent of the Hamiltonian) generates conservative political resistance (Backlash) (Hirschman 1970; Samuelson and Zeckhauser 1988; Kahneman and Tversky 1979) (Figure fig. 18). For the resistance friction R against the rate of change $\dot{J}$, we adopted a "fluid resistance model" proportional to the square of the velocity.

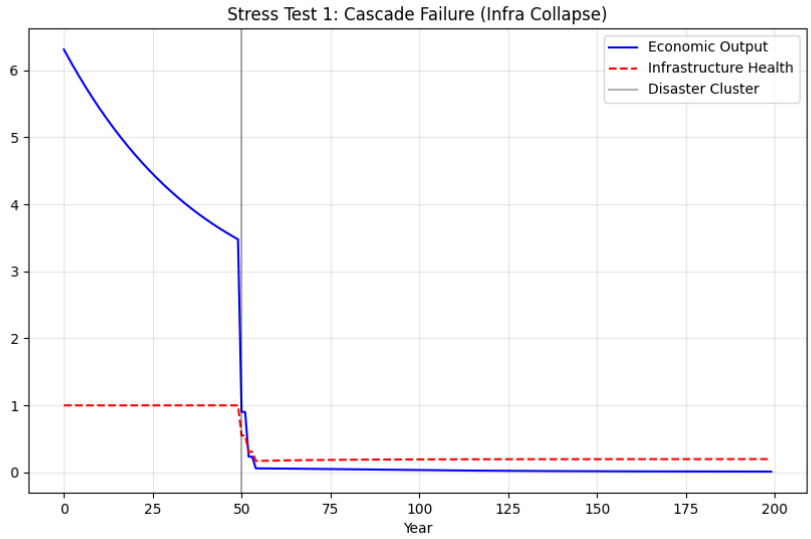

Figure 17: Simulation of cascade failure. When infrastructure resilience falls below a threshold, economic activity stagnates severely.

$$R(\dot{J}) = \mu(\dot{J})^2 \quad (\text{if } R > R_{thresh} \rightarrow \text{Rollback}) \tag{59}$$

While "naive optimization (red line)" is pushed back by the friction coefficient, "adiabatic optimization (green line)" reliably reached the solution at a speed that did not exceed the resistance threshold. This numerically backed the necessity of System 2 (adiabatic intervention).

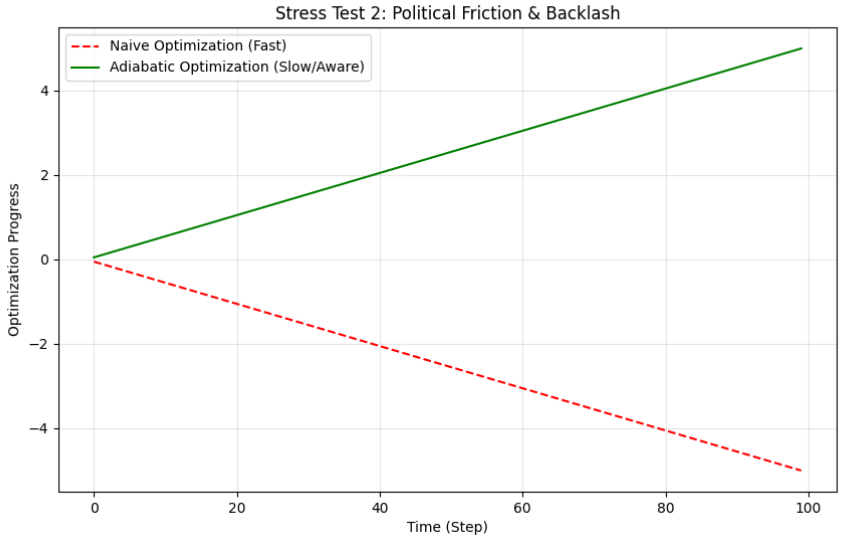

Figure 18: Political friction and optimization path. Only adiabatic processes can avoid resistance.

7.4 Entropy Paradox (Risk of Human Homogenization)

One of the important risks occurs when the system over-optimizes humans (Figure fig. 19). If humans become highly rational (Rationality $r \to 1.0$), diversity (Novelty) is lost, and the innovation coefficient $\mathcal{N}$ is assigned a significantly reduced value.

$$\mathcal{N}(r) = \max\big(0, \min(1, 1 - \xi(r - r_{crit}))\big) \tag{60}$$

In this paper, we call this the "Entropy Paradox." This result shows that there are limits to the "optimization of citizens" through education and nudges (Thaler and Sunstein 2008), and gives a paradoxical suggestion that a certain degree of "irrationality (waste)" must be intentionally protected (Scott 1998; Page 2007).

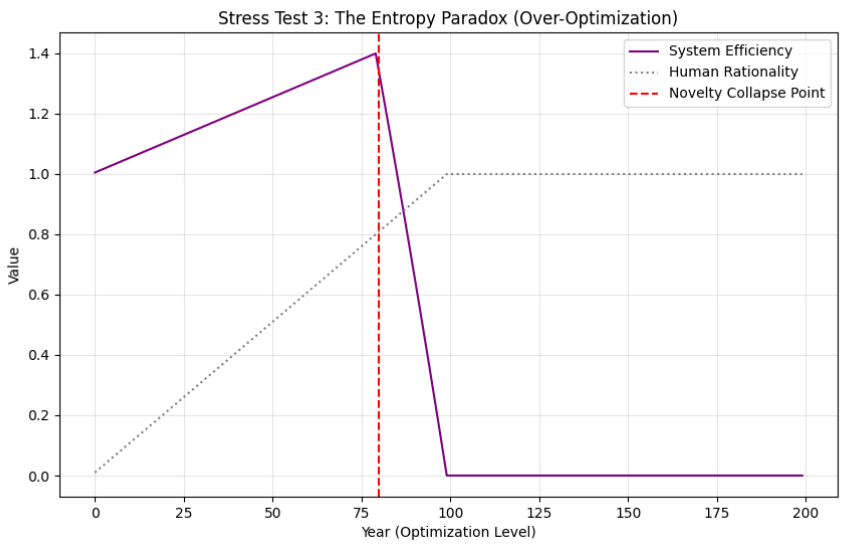

Figure 19: Entropy Paradox. As rationality reaches its extreme, innovation decreases.

7.5 Semantic Hacking Stress Test

In this chapter, "hostility/attack" refers to acts where an attacker intentionally manipulates **reference points** such as inputs (oracles), narratives, and audit indicators to misguide social state transitions and evaluations (it is not a context that describes society or citizens as "enemies").

In a system that introduces a regulated LLM and semantic intervention, **adversarial manipulation** conducted by an attacker against input data, oracle output, narratives (presented text), and audit indicators (proxy indicators) can distort social state transitions (Biggio, Nelson, and Laskov 2012; Szegedy et al. 2014; Goodfellow, Shlens, and Szegedy 2015). Furthermore, when LLMs are ubiquitous in society as proofreading, translation, search, and dialogue UIs, there is a risk that vocabulary/interpretation will be fixed as a "static target" as language operation is normalized and orthographic variations and semantic changes are suppressed (Quiñonero-Candela et al. 2009; Gama et al. 2014). In this case, attackers can easily iteratively optimize induction templates based on past data, and distribution drift type attacks are particularly easy to mimic "natural changes in society."

Under this premise, this system does not depend only on narrative generation (prompt design), but incorporates "lexical rotation-type semantic intervention (semantic versioning / lexical rotation)" as an initial design, which version-controls and updates the reference dictionary and terminology system itself. The aim is not to fix language as a static target for attackers, and the point is to design updates of reference points as institutions and operations, rather than waiting for "naturally occurring language change."

Specifically, (i) the normative layer is fixed by linking legal effects and constraints to immutable Concept IDs, while (ii) labels for citizens (names/abbreviations) and contextual requirements such as application conditions and typical examples are updated at regular cycles. Furthermore, (iii) for old vocabulary, explicit translation (mapping) is required after a compatibility period, visualizing the switching of reference points as a social event. Through this, we make it easier to track and detect distribution drift type attacks, which mimic "natural changes in society" and accumulate over long periods, by dividing them at update boundaries.

Therefore, in this paper, we model semantic hacking into the following three types and stress-test the safety of intervention:

1. **Narrative Poisoning**: Injecting text that makes the social significance of a specific task look exaggerated, biasing the estimation of the external field $\vec{h}_i$.
2. **Goodhartization**: Breaking the correlation between audit indicators (proxy indicators) and true well-being, inducing an apparent decrease in H (Goodhart 1975; Strathern 2000).
3. **Drift Induction**: Artificially generating distribution drift to continuously attack the "near boundary" of constraint Φ_{core}.

As evaluation indicators, we measure (i) constraint violation rate, (ii) ΔH (change in objective function), and (iii) the number of rollback/freeze triggers, confirming that operational layer defenses (audit, freeze, Veto) can safely stop even under attack.

7.5.1 Countermeasures to Semantic Hacking: Three Layers of Audit, Containment, and Fallback

Semantic hacking is difficult to contain with a single measure (e.g., prompt design, lexical rotation). Therefore, in this paper, we drop it into operational specifications as defense in at least the following three layers:

- **Audit Layer (Detect)**: Detect signs of drift/Goodhartization using distribution distance $D(\mathcal{D}_t, \mathcal{D}_{t-k})$ or a decrease in correlation between proxy indicators and proxies for well-being (e.g., ρ_{MW}). These thresholds (θ_D, θ_ρ, etc.) are handled as parameters to be calibrated in operation, rather than being determined as fixed values.
- **Containment Layer (Contain)**: Immediately after detection, apply (i) step clipping (update width limitation), (ii) freeze, and (iii) degradation of the scope of influence (isolation by community unit) to localize damage.
- **Fallback Layer (Recover)**: If convergence is not achieved even with audit/containment, or if the boundary of constraint Φ_{core} is approached, transfer to Analog Fallback (minimum specification of fallback (proposal)) and restore through human trial (System 2) and specification updating (induction).

7.6 Additional Experiments (A–D): Measurement Results under Attack (Adversarial Manipulation) Conditions

To connect the discussion in this chapter to the facts (figures/numerical values) of additional experiments, we organize four additional stress tests using simulation experiments. What is important here is that the

preceding "lexical rotation (semantic versioning / lexical rotation)" is a **design countermeasure against the same threat domain as Additional Experiment A (Semantic Hacking)**, and its purpose is different from Additional Experiments B–D. The corresponding relationships are clarified as follows:

- **Experiment A (Semantic Hacking)**: Related in the position of making it easier to track and detect long-term drift accumulation "by dividing it" by switching reference points (dictionaries/vocabulary) at update boundaries against Poison/Goodhart/Drift handled in this section.
- **Experiment B (zk-benchmark)**: Evaluation of performance (TPS/latency) related to proof generation and aggregation, and **not directly related** to resilience against semantic hacking (though it is separately important as an operational cost constraint).
- **Experiment C (Oracle)**: Evaluation of error reduction by external input noise and fallback, related more as a problem of **input quality and measurement robustness** than semantics.
- **Experiment D (Trust Bottleneck)**: Confirmation that social acceptance time T_{accept} becomes dominant, indirectly related to the discussion of **understanding and acceptance costs** brought about by institutional designs such as terminology update events.

7.6.1 Experiment A: Semantic Hacking

In combinations of attack strength (Poison/Goodhart/Drift) and defense conditions (Defense), we observed average Hamiltonian (Avg_H), final Hamiltonian (Final_H), constraint violations (Violations), and Veto triggers (Vetoes). Under Defense conditions, Vetoes were frequently triggered (e.g., Vetoes=175) against Poison (Strong), and while constraint violations themselves were suppressed, cases were confirmed where many constraint violations were observed (Violations=266) even though Vetoes were not triggered under conditions with strong drift (Drift (Strong)).

7.6.2 Extended Specification of Analog Fallback: Triggering and Recovery under Semantic Hacking

The "minimum specification of fallback (proposal)" described in this paper is introduced primarily as a "last resort" for oracle errors and dispute processing. On the other hand, the results of Experiment A show that there can be situations where **Vetoes do not trigger but constraint violations accumulate**, such as Drift (Strong), in addition to situations where Vetoes trigger frequently and the system **can stop**, such as Poison (Strong).

Therefore, in operation under semantic hacking, "trigger conditions," "preservation of evidence," and "reflection of processing results" included in the minimum specification of fallback (proposal) must be concretized according to the failure modes observed in stress tests. Here, "normal operation" refers to a state where automated updates are permitted, and "safe mode" refers to a state where damage is localized by step clipping / freeze / containment. Note that thresholds ($\epsilon, \theta_D, \theta_\rho, \theta_{viol}$, etc.) and window widths (k, M, etc.) are handled as parameters to be calibrated by operation, not determined as fixed values.

As **trigger conditions (example)**, the system shifts from normal operation to safe mode if any of the following conditions are met, and if improvement is still not achieved, it transfers to analog fallback:

- **Exhaustion of constraint margin**: $\min \mathrm{margin}(\Phi_{core}) < \epsilon$.
- **Drift warning**: $D(\mathcal{D}_t, \mathcal{D}_{t-k}) > \theta_D$ for M consecutive times.
- **Goodhart warning**: $\rho_{MW} < \theta_\rho$ (decrease in correlation between proxy indicator M and well-being proxy W).

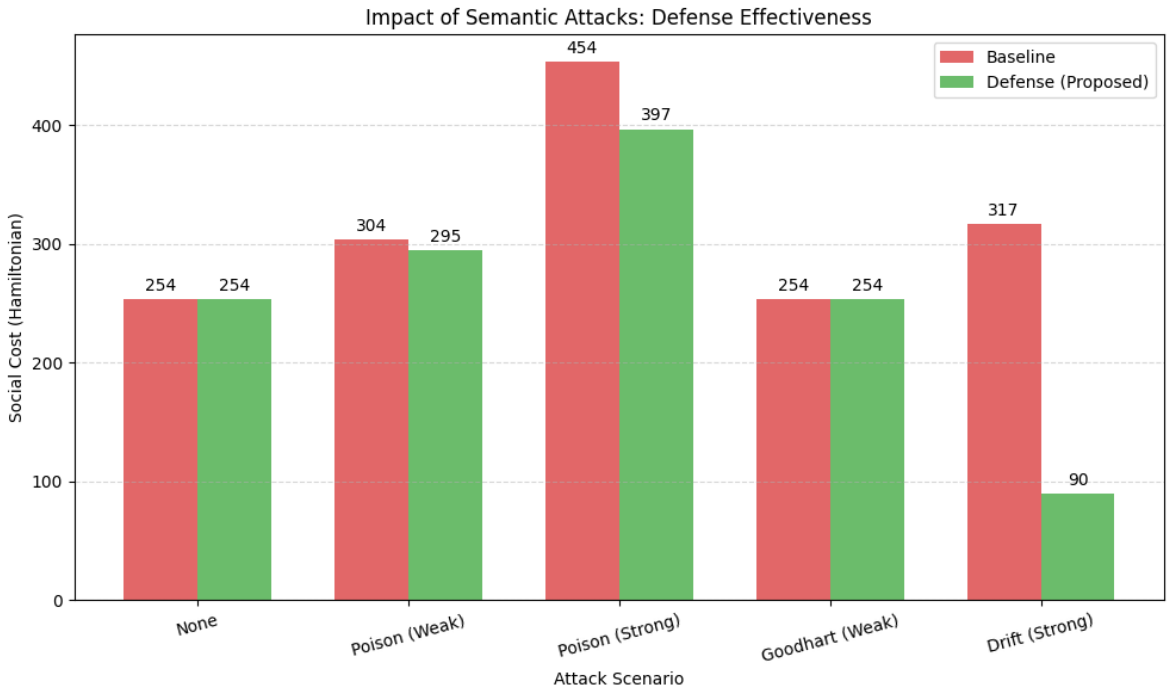

Figure 20: Results of Additional Experiment A (Semantic Hacking). Comparing the behaviors of Avg_H/Final_H, Violations, and Vetoes under attack type/intensity and defense conditions (Defense).

- **Veto non-trigger x Violation increase**: Violations/$\Delta t > \theta_{viol}$ and $N_{veto} = 0$ (corresponding to Drift mimicry).

During fallback, (i) automated updates are stopped. For **preservation of evidence**, logs, sensor data, and signature sequences at the time of fallback triggering are hashed and fixed to the ledger, and preserved in a state where they cannot be tampered with later. For **reflection of processing results**, the conclusion of the human trial is reflected as (a) approving a one-time state transition as exception processing, or (b) updating the specifications (constraints, weights, oracle requirements), inducing identical cases into automated processing in the future.

7.6.3 Experiment B: zk-benchmark (Batch size and TPS/latency)

The increase in batch size brings an improvement in theoretical throughput (TPS), but increases latency derived from aggregation and proof generation (proving). In this additional benchmark, it was confirmed that while TPS(Est) reaches about 6,400 with a batch size of 64 or more, latency tends to increase to the order of several hundred ms to seconds (even assuming 64 cores).

7.6.4 Experiment C: Oracle Errors and Risk Reduction by Fallback

As the noise level of the external input (oracle) rises, errors increase, but errors are significantly reduced by introducing fallback. For example, at NoiseLevel=0.01, Errors(NoFallback)=10 vs Errors(Fallback)=2 (RiskReduction=80%), and a certain risk reduction was observed. On the other hand, the number of fallback triggers (FallbackCount) increases (57 under the same conditions), accompanied by operational costs

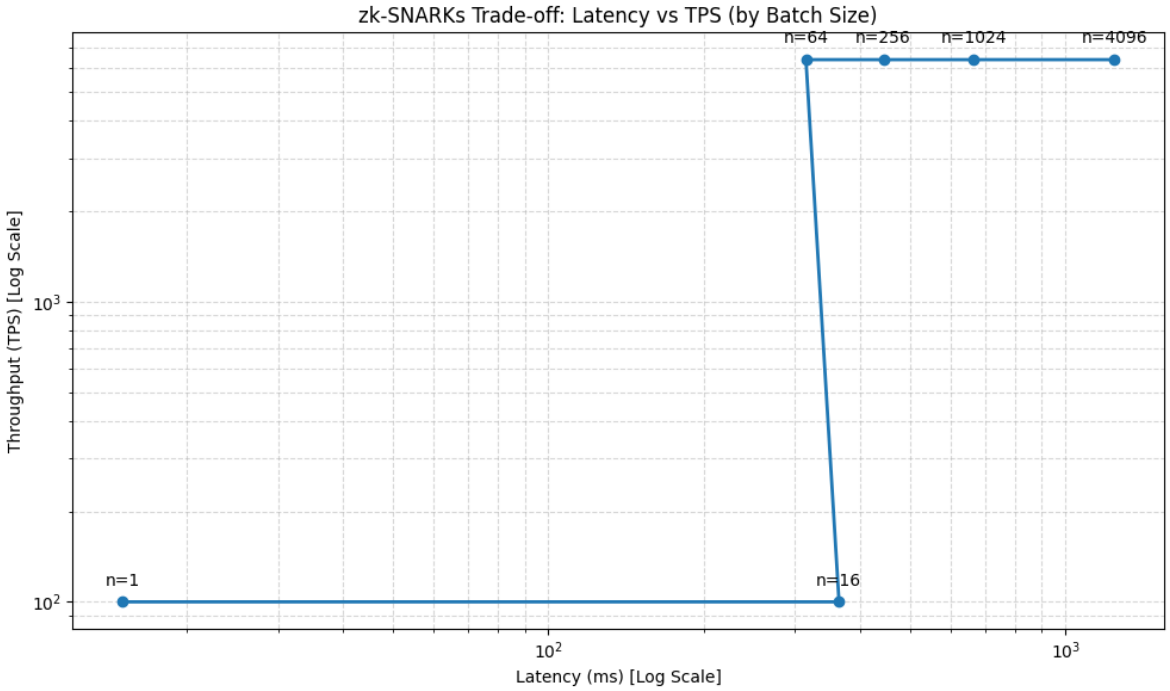

Figure 21: Results of Additional Experiment B (zk-benchmark). Visualizing the trade-off of TPS/latency by Batch size.

(frequency). In this sense, fallback is positioned as an application example of analog fallback to return the system to the safe side in times of anomaly, rather than "optimization in normal times."

7.6.5 Experiment D: Trust Bottleneck (T_{accept} dominant region)

Even if computing time T_{calc} is minimal, as long as social acceptance time T_{accept} is dominant, the total resolution time T_{total} will not be shortened. In additional experiments, it was confirmed that in areas where trust τ is small (e.g., $\tau = 0.1$), T_{accept} increases significantly ($T_{total} \approx 1000$), and the effect of T_{calc} improvement becomes limited.

The above is an attempt to quantitatively show that the bottleneck of T_{accept} (social acceptance) can be dominant, and T_{total} is not shortened only by improving T_{calc}, against the leap in logic that improvement in computing speed shortens the social resolution time. Therefore, the prerequisite for the establishment of this system lies in the design of consensus-building infrastructure (Trust Protocol) that pulls up the trust coefficient τ, rather than the reinforcement of computing resources.

7.6.6 Countermeasures for Trust Bottleneck: Shortening Design of Acceptance Time T_{accept}

Experiment D showed that improvement in computing time T_{calc} alone does not shorten the overall resolution time T_{total}, and social acceptance time T_{accept} can be dominant. Therefore, in operational design, it is necessary to treat T_{accept} not as an "exogenously given delay," but as a design variable to be managed as an institutional specification. We propose at least the following:

- **Two layers of public periods (L1/L2)**: Layer 2 (Policy Parameters) has a standard public period

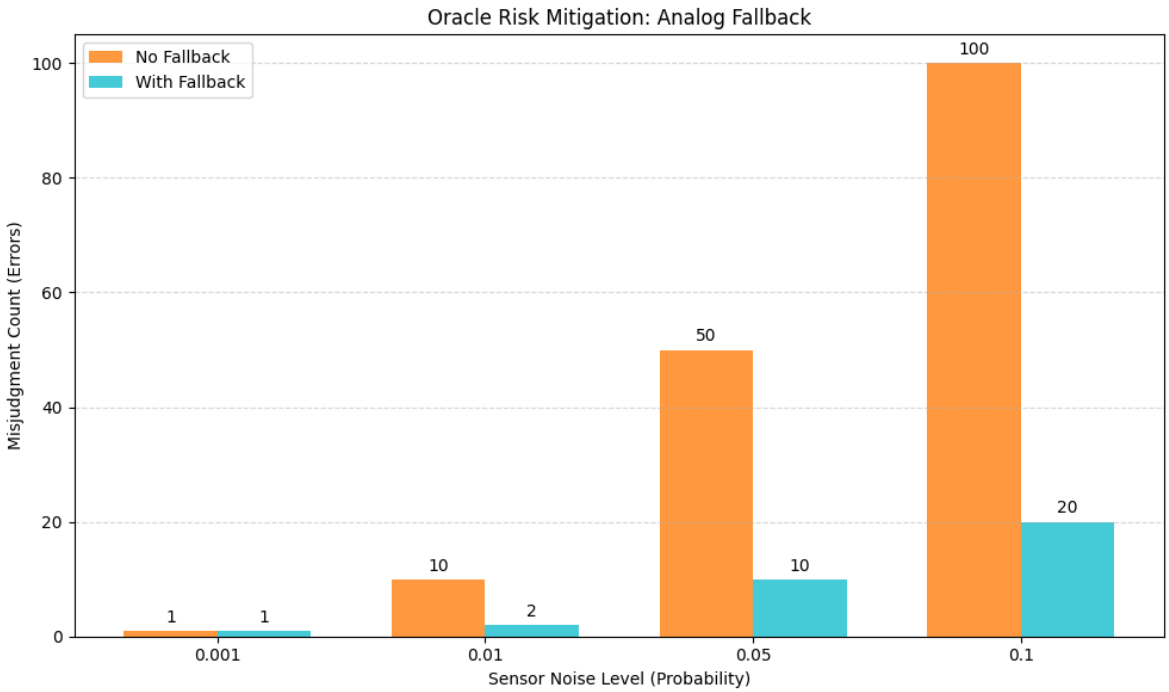

Figure 22: Results of Additional Experiment C (oracle). Showing errors against noise levels and risk reduction (and trigger frequency) by fallback.

Δt_{L2}, and Layer 1 (Constitutional Core) has $\Delta t_{L1} \gg \Delta t_{L2}$, separating acceptance processes according to the severity of changes.

- **Prohibition of "Borrowing" from Acceptance**: Explicitly stating that high-speed optimization (small T_{calc}) cannot substitute for a lack of acceptance, and not omitting disclosure, objection filing, and explanation requests.
- **Observability**: Recording opposition rate / objection rate, the number of re-proposals, the number of explanation requests, and audit NG rate as metrics to monitor whether acceptance delays are structured.
- **Stepwise Introduction (Canary)**: Instead of simultaneous total application, deploy stepwise starting from sub-communities with small scopes of influence, localizing rollback costs in case of failure.

7.6.7 Auto-Immune Paradox

The systemic behavior where optimization leads to the suppression of human-derived stochastic variation is illustrated in Figure fig. 24. This simulation demonstrates that... The red line is when the order parameter of the system is brought close to 1.0 (complete control), falling into the "risk of excessive control" where the innovation coefficient decreases and economic value is greatly lost. In contrast, the green line is when control is intentionally stopped at 0.7, and the remaining 0.3 is protected as a "Wildness Reservation." In this case, economic value, which is the product of order (efficiency) and chaos (innovation), is maximized. This numerically supports that CSCT aims not at "complete taming of humanity" but at "strategic protection of wildness."

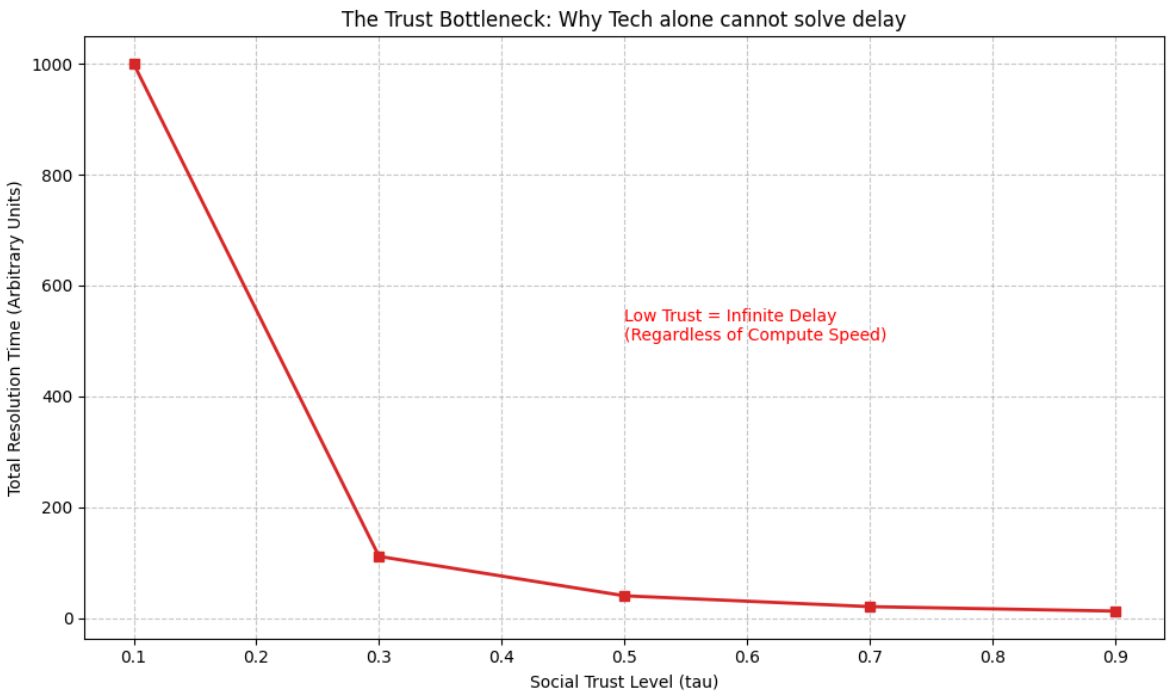

Figure 23: Results of Additional Experiment D (trust bottleneck). In areas where τ is low, T_{accept} dominates T_{total}.

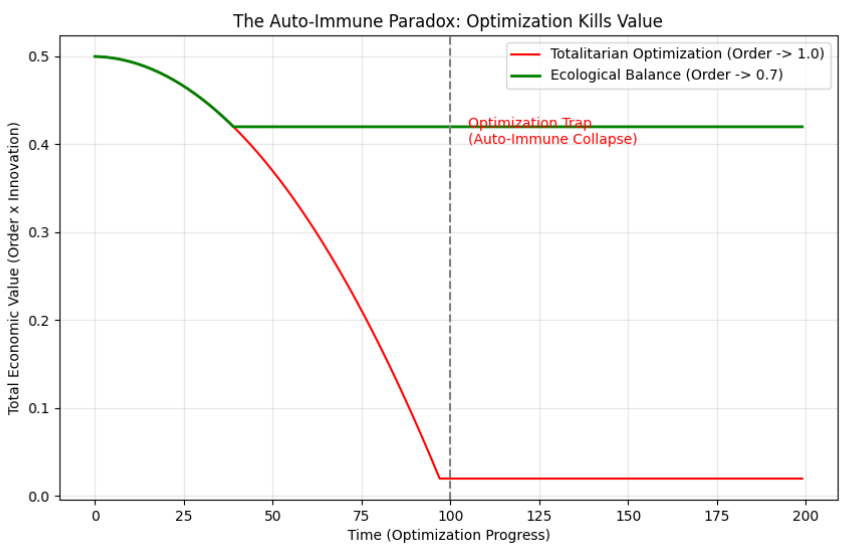

Figure 24: Auto-Immune Paradox. Excessive optimization (red) depletes "noise (humanity)," which is the energy source of the system. Only moderate control (green) is sustainable.

7.7 Reproducibility and Ablation Plan

To show that the stress test results in this chapter do not excessively depend on specific assumptions or parameters, we clarify the following reproducibility information and ablation (mechanism removal).

7.7.1 Reproducibility (Minimum Recorded Items)

- **Random seed**: Seeds used for Figures fig. 17, fig. 18, fig. 19 (and the semantic hacking stress test).
- **Default parameters**: Default values of $(A, \alpha, \beta, \delta, \eta, \mu, R_{thresh}, \xi, r_{crit})$.
- **Attack scenario definition**: Generation rules of poisoning prompts, operations of Goodhartization (tampering methods of proxy indicators), and procedures for drift induction.
- **Update laws**: Corresponding equations in the text (simultaneous equations, resistance model, $\mathcal{N}(r)$ in this chapter) and their application order.
- **Evaluation indicators**: Economic output P_t, infrastructure health I_t, the number of rollbacks/freezes, time average of $\mathcal{N}$, constraint violation rate, etc.

7.7.2 Ablation (Which mechanisms are working)

1. **Without Redundancy**: Decrease the upper limit of the infrastructure recovery term (K_{max}) and confirm the threshold dependence of cascade failure.
2. **Without Adiabatic Condition**: Remove the rate-of-change constraint and confirm the increase in rollback frequency due to $R(\dot{J}) > R_{thresh}$.
3. **Without Wildness Protection**: Confirm that $\mathcal{N}$ depletion is structurally unavoidable by forcing $r \to 1.0$ (or decreasing r_{crit}).

8 Discussion and Conclusion

In this chapter, we discuss the operational risks and ethical issues of the proposed system and present the conclusions of this paper.

8.1 Scope Limitation and Level of Claims

The Hamiltonian used in this paper does not claim strict isomorphism with all processes of actual society, but is introduced as a **constructive objective function to consistently describe trade-offs in institutional design**. That is, the role of the Hamiltonian is to provide a design language that makes efficiency, constraint compliance, and trust indicators comparable on the same coordinate system.

For this reason, the quantitative results of this paper (long-term MAS, efficiency rate, comparative scenarios) are positioned not as definitive predictions of policy effects, but as **exploratory simulation results** dependent on assumptions. For reproducible empirical confirmation, (i) fixation of variable definitions, (ii) data connection ensuring external validity, and (iii) presentation of sensitivity analysis and falsifiability are additionally required.

8.2 Preparing for Quantum Impact

One of the major risks of the proposed model is the vulnerability of the cryptographic foundation accompanying the progress of quantum computers. If current public-key cryptography is broken, "high-level trust" based on task ownership and evaluation history could be damaged in a short period.

In response, the introduction of "Quantum-resistant cryptography (e.g., Lattice-based cryptography) (Regev 2009)" in the aforementioned Latent Torus concept is an essential condition. Furthermore, in system operation, it is premised that a mechanism is constructed to absorb and distribute social fluctuations (shocks) with complexity that cannot be processed by conventional computers by treating the entire social system as one huge many-body problem and performing calculations using optimization algorithms on quantum computers (Kadowaki and Nishimori 1998; Farhi, Goldstone, and Gutmann 2014).

8.3 Difficulty of Counterfactual Analysis and Structural Break

As an approach to verify the effectiveness of this theory, a "counterfactual" evaluation of "what would have happened if this system had existed" against past economic crises (e.g., the 2008 global financial crisis or the 2020 pandemic) appears attractive. However, since CSCT / Latent Torus is not a mere change of policy parameters but **institutional intervention that recreates the Data Generation Process (DGP) itself**, counterfactuals by extrapolation of existing statistics will collapse with high probability. In this paper, we organize that difficulty as a "Structural Break" into the following three points:

1. **Break in Unit and Scale (Change in Measurement System)**: In existing society, indicators such as labor, income, unemployment, and GDP are defined based on fiat currency and employment contracts. On the other hand, in this system, the unit of value shifts to tasks (and τ), and subjects are not limited to humans, including LLMs, robots, and automated institutions (protocols). At this time, even with statistics of the same name, "what is being measured" changes, so simple comparison with past series becomes **conceptual inconsistency** rather than **measurement error**.

2. **Mutual Interference and Reflexivity (Intervention Changes Behavior)**: Since this system continuously intervenes in behavioral preferences through pricing, allocation, and narrative generation, the assumption that individual intervention effects accumulate independently (the so-called independence/non-interference premise) is difficult to hold. Furthermore, since there is feedback where what the institution "predicted" changes behavior and those changes return to the next optimization input, the object of evaluation is not a static policy but an **adaptive control system**. We consider that an attempt to quantitatively evaluate nudges is necessary to clarify this.

3. **Non-invariance of Structural Parameters (Lucas Critique)**: Past data is generated in a parameter space of centralized currency issuance, human-centered labor markets, and limited information observation. In contrast, CSCT assumes distributed consensus, cryptographic auditing, and automated optimization, plus "allowable constraints" and "objective functions" can be operationally updated (versioned). Therefore, structural equations (demand functions, labor supply, etc.) fitted to past data are not preserved after policy changes, and the Lucas Critique (Lucas 1976) becomes manifest **in a stronger form**.

From the above, verification of this theory must be conducted in three layers: (i) identification of invariants at the mechanism level, (ii) quasi-experimental evaluation in sandboxes/stepwise introduction, and (iii) generative social simulation via Multi-Agent Simulation (MAS). In particular, MAS is positioned merely as an "experimental environment" to measure robustness as a distribution by executing counterfactuals under the same institution (ON/OFF of intervention, strength of constraints, difference in update cycles) many times.

8.4 Ethical and Legal Issues

Social implementation of "absolute trust" involves ethical issues such as loss of privacy (Nissenbaum 2010) and black-boxing by algorithms (Pasquale 2015). However, in this study, we proposed a solution called "Transparency Stratification."

That is, what citizens seek may be "fairness of transaction results (backend transparency)" rather than the transparency of the optimization process itself (Latent Torus). If the Latent Torus functions sufficiently and continues to reduce social friction to some extent, people's motivation to track internal details may relatively weaken. However, it is important not to set the decline of transparency requirements as a desirable goal itself, but to design which layer of transparency is necessary while maintaining mechanisms for auditing, objection filing, and accountability.

This is a context where we expect the existence of the Latent Torus and the algorithms implemented in society to become ubiquitous (environment-ified), just as people live every day without being conscious of the laws of physics, details of the human body, or existing authority.

8.5 Remaining Challenges and Risk Management

We describe technical and institutional challenges that still remain toward social implementation of this system.

8.5.1 Probabilistic Limits of the Oracle Problem and Analog Fallback

While objective truth certification by sensors and multi-sig (Section 4.1.7) targets high accuracy, errors can remain due to measurement errors, attacks, collusion, etc. (numbers depend on operational conditions, environment, and attack models). This is also a limit known as the "oracle problem" of smart contracts (Egelund-M"uller et al. 2017).

In response, the system needs to be equipped with an "Analog Fallback" mechanism. If probabilistic consensus building falls below a prescribed threshold (if the dispute term $H_{dispute}$ does not converge to a local minimum), the case is automatically escalated to a "human judge (System 2)." Rather than aiming for full automation, holding this "final judgment by humans" as the last resort ensures the robustness of the system.

Minimum Specification of Fallback (Proposal): To ensure that analog fallback functions as a "safety valve" rather than an "exception," we specify the following at minimum:

- **Trigger Conditions (Example)**: (i) Contradiction between sensors, (ii) inconsistency in testimony/evaluation graphs, (iii) high suspicion of tampering by anomaly detection, (iv) when the fulfillment margin of Φ_{core} approaches less than a threshold.
- **Preservation of Evidence**: Logs, sensor data, and signature sequences at the time of fallback triggering are hashed and fixed to the ledger, and preserved in a state where they cannot be tampered with later (while minimizing viewing permissions).
- **Reflection of Processing Results**: The conclusion of the human trial is reflected as (a) approving a one-time state transition as exception processing, or (b) updating the specifications (constraints, weights, oracle requirements), "inducing" identical cases into automated processing in the future.

Thus, fallback should be designed not as a power where "humans overwrite the algorithm," but as a learning circuit that "returns uncertainty to specification updates."

8.5.2 Neutrality of Semantic Intervention and Model Governance

In the narrative generation process by the regulated LLM, the risk of specific ideologies or biases being mixed in (Semantic Bias) is unavoidable. As a countermeasure, it is required to place the Embedding Model itself under "open source governance." Using Adversarial Training, it acquires robustness to output most neutral vectors against attack models with various biases. Also, a system where the weight parameters of the model itself are managed and updated by a DAO (Decentralized Autonomous Organization) is essential.

Countermeasures to Semantic Hacking (Outline): Managed LLMs have attack surfaces such as (i) input narrative poisoning, (ii) Goodhartization of proxy indicators, and (iii) constraint deviation due to distribution drift. Therefore, three-layer defense is required at minimum:

- **Specification Layer**: Explicitly stating formal constraints of Φ_{core} (anti-discrimination, due process, etc.) and excluding searches that cause violations from the state space.
- **Learning Layer**: Detecting early the collapse of correlation between evaluation functions and reality through drift detection, counterfactual tests, and anomaly detection.
- **Operational Layer**: Stopping irreversible runaway of system updates through red teaming, update freezing, and audit committees (and Veto).

In future demonstrations, changes in H and constraint violation rates when adversarial inputs are given must be evaluated in a reproducible manner.

8.5.3 Uneven Distribution of Computing Resources and Edge-First Strategy

Since real-time behavioral sensing and advanced cryptographic computation require powerful infrastructure, there is a risk of leading to centralization of power (Centralization of Compute) to wealthy classes and large corporations with computing resources in the early stage. To prevent this, the architecture should be designed as "Edge-First." Signature processing of σ_{truth} is completed within individual smartphones and IoT devices, and only zero-knowledge proofs are sent to the central server. Thus, it becomes possible to optimize the entire system while keeping data sovereignty in the hands of individuals.

8.6 Conclusion: Absolute Trust and the Future of Humanity

In this study, we reinterpreted the superfluidity of tasks not merely as a means of economic efficiency, but as an opportunity to restructure the social system itself, and presented a theoretical framework using Hamiltonian as a design language.

In conclusion, "tokenized task transactions" and the "Grand Unified Algorithm" developed by incorporating them are useful as design hypotheses for examining the "financialization of meaning" beyond the liquefaction of money and the computability of consensus building. By implementing the black box of "absolute trust" called the Latent Torus as the social kernel, part of the cognitive load associated with "governance" can be reduced. On the other hand, the design of the institution's target function, constraints, and update procedures (and civil control over them) remains indispensable, and the claim of this paper is not "the exit of humans" but the "redistribution of the judgment domain that humans should handle."

Furthermore, even in the case of ultimate labor saving by AI (a state where many routine and semi-routine tasks are automated and the density of human involvement is significantly reduced), it is significant that CSCT can design the transition from existing institutions not as a disruption but as a gradual connection. That is, by holding the procedures, audits, and remedies of traditional institutions as fallbacks while incrementally expanding the provable automatic execution domain, there is a possibility to avoid sudden institutional voids or large-scale coordination failures, relatively leveling transition costs (AbdelStark 2026).

When this theory is applied and implemented in society, it may prompt a re-examination of "Humanism" since the Renaissance. At the very least, it can force a reconsideration of the division of roles in the design, operation, and supervision of institutional operations by people. However, that does not necessarily mean that the public will is disconnected from the operation of the social system.

Furthermore, from the perspective of expanding application, the concept of a supranational accounting unit, "Bancor," proposed by Keynes in the blueprint for the ICU (International Clearing Union) is suggestive. If the objective function and constraint system of the Grand Unified Algorithm and CSCT can be expanded to treat international trade imbalances as a "system distortion"—rather than being closed to domestic task markets—we might incorporate trade balance correction as a design theory. Implementing adjustment rules via smart contracts that satisfy sovereignty and transparency requirements could bring an "International Latent Torus" into view as a future research subject.

At the same time, even in a scenario where the population surges and interaction density rises discontinuously, CSCT can maintain resilience as an ultra-multi-body optimization algorithm. The priority here is not "strengthened control" but a design that suppresses constraint violations in an observable manner while expanding the parameter space handled across optimization, verification, and auditing. While rapid population growth generally increases the risk of social collapse, a framework equipped with specification recursion, provable fallbacks, and decentralized execution has the potential to mitigate these risks (AbdelStark 2026).

We hope this paper serves as a design proposal for the potential structural changes in labor markets and the impact of quantum computation. It is essential to continue searching for institutional alternatives and verifiability, rather than dismissing technological tools due to their complexity.

Limitations of the Numerical Framework

While the simulations presented in this study provide a visual and numerical demonstration of the Social Hamiltonian dynamics, they are inherently limited by the high degree of idealization required to model human-AI cognitive feedback loops. The semantic intervention functions and cognitive response parameters represent simplified, exploratory proxies rather than empirically derived metrics. Consequently, the results should be interpreted as a *numerical thought experiment* intended to verify the logical consistency of the theoretical framework. The "Grand Unified Algorithm" behavior observed here is a product of these idealized conditions, serving as a preliminary proof-of-concept for future empirical verification in real-world decentralized systems.

Note on Translation

The English version of this research paper was generated through a translation process from the original Japanese manuscript utilizing Large Language Models (LLMs). While rigorous efforts have been made to maintain scientific accuracy and terminological consistency, the original author remains the final authority for the interpretation of the concepts and nuances discussed herein.

Data and Code Availability

The Python simulation source code, C-based optimization modules, and raw benchmarking logs supporting the findings of this study are open-source under the Apache License 2.0. The complete `simulation_audit_package` (available at https://github.com/BANAN-monguram/Social-Hamiltonian), including a detailed reproducibility guide and environment configurations, is hosted on GitHub. Researchers are invited to clone the repository to reproduce the Phase 1–8 simulations and verify the stability limits of the proposed Hamiltonian dynamics. Please note that these numerical results represent idealized theoretical explorations rather than absolute performance benchmarks.

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